

MTD / PhD / EngD / MEng ·

SEPTEMBER – DECEMBER 2026

Innovation by Design (Week 3)

14 WEEKS

4 MILESTONES

5–50 UNITS

1 SALES
GALA/FINAL
PRESENTATION

Define: problem scoping; benchmarking existing alternatives; needs synthesis; product-service-system requirements; early customer and venture hypothesis.

Faculty Lead Dr Ark Kang · Co-Leads Raghavan Nagarajan, Huang Shaoying

Main agenda: Define, Clustering, Ranking Needs, Prepare for Week 4 Presentation

INDICATIVE SCHEDULE

Fourteen weeks

1	14–18 Sep	Introduction, teams, AI toolchain	Journal 1	8	2–6 Nov	Milestone 2 · Ideas and Feedback	Presentation
2	21–25 Sep	Discover: interviews, AI research	Journal 2	9	9–13 Nov	Deliver: prototyping and CAD	Journal 7
3	28 Sep – 2 Oct	Define: scoping and benchmarking	Journal 3 · M1 due	10	16–20 Nov	DFMA, costing, unit economics	Journal 8 · M3 due
4	5–9 Oct	Milestone 1 · Venture Thread	Presentation	11	23–27 Nov	Milestone 3 · Venture Readiness	Presentation
5	12–16 Oct	Develop: ideation, generative AI	Journal 4	12	30 Nov – 4 Dec	Manufacture and launch prep	Journal 9
6	19–23 Oct	Develop: selection and price	Journal 5 · M2 due	13	7–11 Dec	Launch, sell, fulfil, iterate	Journal 10 · M4 due
7	26–30 Oct	Break week · synthesis of feedback	Journal 6	14	14–18 Dec	Milestone 4 · Exhibition & Gala	Presentation

Highlighted weeks are presentation weeks. Week 7 is a break week

AGENDA: Discover, Team Dynamics, Refine Approach for Discover



Define

understanding customer needs;
evidence-based customer interviews;
observations; AI-assisted desk
research and needs synthesis.

Share and reflect insights from your
discover phase, think of how your
questions and approach can be
refined.



Clustering Insights And needs from interviews. Form a team HMW statement

Fill in the Miro Board for your team
HMW statement/Design Direction.
Who is the user? Why the problem
matters?



Sharing of Rubrics for Week 4

Giving team time to work on the Week
4 Presentation. Stakeholder
engagement plan, needs,
requirements, segment, value
proposition

Miro Announcement/ Flexibility of Digital Repo

- We will allow a digital repository of their preference to document all their findings, studies and inferences...." This link should be shared with us during every milestone.
- Your efforts from week 1 to 3 will still be accounted for You all can still continue to use miro if your team prefer them

Following up from Week 2

- From their initial interviews, each student should have their own HMW statement. That should bring about five different topics, each with its relevant user needs, to the team.
- Define phase:
- Team HMW: Decide on and formulate the team's HMW statement(s).
- Cluster needs: Group the user needs under their HMW statements.
- Attempt to translate needs to specifications
- Journey map: Build a journey map around the team's HMW statement.
- Conduct follow-up interviews, and pitch to or role-play with the people they spoke to. This should help them prioritize needs and define the functional requirements for developing their PSS further.
- Development should be driven by what users and customers actually ask for, not by what the team assumes is good.
- Add HMW statements and photos of this work on the miro board

Team Presentation Week 4 – Rubrics Discover-Define

Milestone 1: Discover and Define (Week 4 presentation, 15%)

Criterion	Poor	Good	Excellent
Problem opportunity and scope (25 points)	0–10: HMW statement/Design Goal/Problem is too narrow, too broad, solution-led, or infeasible within the course.	11–18: HMW statement/Design Goal/Problem is feasible and appropriately scoped, with a plausible target user.	19–25: HMW statement/Design Goal/Problem is important, well-bounded, feasible, and clearly connected actual users or market opportunity.
User and stakeholder research (35 points)	0–15: Little or no direct engagement with relevant users; findings are assumptions or isolated anecdotes.	16–26: Relevant interviews, observations, or fieldwork are conducted, with some analysis and learning.	27–35: Strong, ethical, and repeated engagement with actual/relevant stakeholders generates evidence-based insights. The team clearly shows how feedback and design processes changed its understanding of the problem.
Existing-solution and market landscape (20 points)	0–8: Few existing alternatives are identified, or comparisons are superficial.	9–15: Major alternatives are identified and compared, although analysis may lack depth or clear criteria.	16–20: Thorough and critical comparison of direct, indirect, and non-consumption alternatives using meaningful criteria, evidence, and clear identification of opportunity gaps.
Needs, requirements and early venture hypothesis (20 points)	0–8: Requirements are missing, unclear, or replaced by an unjustified proposed solution.	9–15: Needs and preliminary requirements are identified, with an initial target customer and value proposition.	16–20: Clear, traceable requirements are derived from actual user needs research and benchmarking (data sources, comparison metrics, KPIs). The team articulates a credible early-adopter segment, customer problem, value proposition, and willingness-to-pay hypothesis.

Presentation Guidelines

- Strictly 7 mins per team
- 5 mins Q & A allocated per team. TAs will time and stop you at 7:00.
- At least two members must present in each round.
- Speakers must rotate between rounds. If A and B present in milestone 1, then C, D and E present in milestone 2. Every member should present at least twice.
- Label Class – e.g Mon 3pm_Milestone 1_Team_(name)
- Submit final decks within 48 hours of their presentations (to further refine their slides based on our feedback)
- Late submissions penalised 10%
- Format: PPT
- Every member must be able to **answer questions**, regardless of who presented. Judges may direct a question to a specific person.

Presentation Content

- Development of HMW statements, reasons for choice, final HMW statement/ Design Goal/Problem
- Stakeholder Engagement: *Key insights* from user research and interviews, observations.
- Key categories of user needs with breakdown and prioritisation
- Journey map – present key insights from the process (not read out everything)

Presentation Content – 1. HMW Statement (25 marks)

- **One HMW statement on screen** — phrased as "How might we [action] for [specific user] so that [outcome]?"
- **Who the user is, specifically** — not "students" but "first-year international students in their first 6 weeks"
- **Why this problem matters** — size of the affected group, frequency, cost or consequence of it going unsolved, with a source
- **Scope boundaries** — state explicitly what is *in* scope and what you deliberately excluded, and why
- **Market or opportunity link** — the commercial or institutional opening this problem creates
- **Avoid:** a solution disguised as a problem ("How might we build an app that...")

Presentation Content – 2. Stakeholder Engagement (35 marks)

- **Engagement table** - who you spoke to, their role, how many, method (interview, observation, co-design), and when
- **Evidence of repeated contact** - show at least two rounds with the *same* stakeholders, not one-off interviews
- **Range of stakeholders** - primary users plus adjacent ones (payers, gatekeepers, operators, non-adopters)
- **Direct evidence** - 2–3 verbatim user quotes, photos or observation artifacts (AEIOU notes, journey maps)
- **Raw insight** → **reframed insight** - show your "before" assumption and your "after" understanding side by side
- **A pivot with a cause** - name one specific piece of feedback and what it changed in your problem definition or design.
- **Iteration trail** - a simple timeline showing Round 1 → learning → Round 2 → learning → current state
- **Avoid:** claiming "we spoke to many users" without numbers, dates or artifacts

Presentation Content – 3. Benchmarking and Gaps (20 marks)

- Three categories explicitly labelled:
 - Direct competitors (same solution, same problem)
 - Indirect substitutes (different solution, same job)
 - Non-consumption (what people do to work around it manually)
- A comparison matrix — alternatives as rows, your evaluation criteria as columns
- Criteria justified by user needs, not convenience — say why each criterion matters to the user
- Evidence per cell — pricing pages, feature documentation, app store reviews, user testimony. Cite sources
- Honest assessment — name where an existing alternative beats you
- The opportunity gap stated in one sentence — what no existing option does, and why that gap persists
- Avoid: a matrix of green ticks in your own column and red crosses everywhere else

Presentation Content – 4. Needs, Requirements, Segment & Value Proposition (20 marks)

- **Traceability table** - each requirement mapped back to its source: Requirement ← user need ← evidence (interview #, benchmark, data source)
- **Requirements split** into must-have, should-have, and out-of-scope
- **KPIs with numbers** - what you will measure, the baseline, and the target (e.g. "task time from 12 min → under 4 min").
- **Early-adopter segment, narrowly defined** - who feels the pain first and hardest, and why they will tolerate a rough version 1 prototype
- **Customer problem statement** for that segment specifically
- **Value proposition in one line** - for [segment] who [problem], our [solution] delivers [benefit], unlike [main alternative]
- **Willingness-to-pay hypothesis** - the price point, *who pays* (user, institution, agency), and the evidence behind it: what people currently pay for the alternative
- **How you will test WTP (Willingness-to-pay) next** - pre-orders, letters of intent, pilot agreements
- **Avoid:** "everyone could use this" - an unbounded segment scores poorly here

Presentation Content – 4. Needs, Requirements, Segment & Value Proposition (20 marks)

- **Traceability table** - each requirement mapped back to its source: Requirement ← user need ← evidence (interview #, benchmark, data source)
- **Requirements split** into must-have, should-have, and out-of-scope
- **KPIs with numbers** - what you will measure, the baseline, and the target (e.g. "task time from 12 min → under 4 min")
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Define Phase



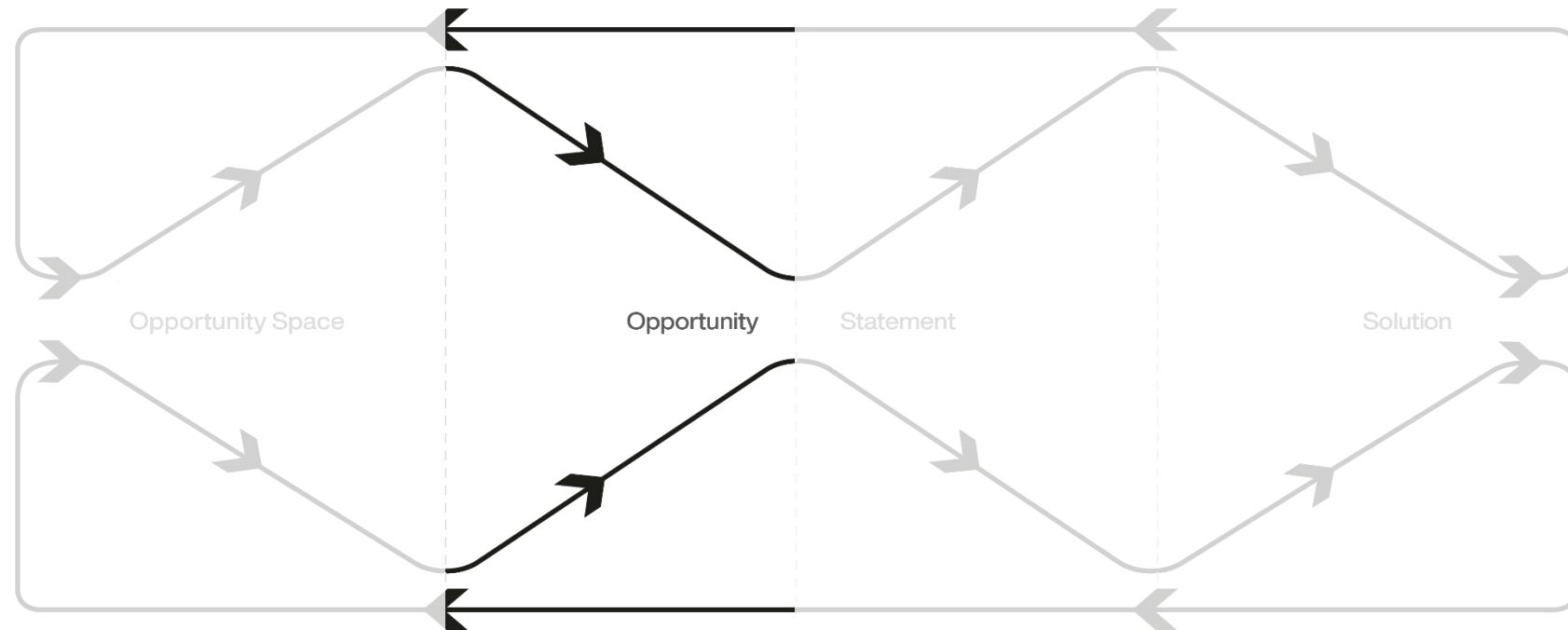
Across disciplines, problems are identified, defined and initiated differently.

In some areas, the process is viewed as '**messy**' and undetermined, others follow more systematic approaches.

In all cases, information gathering and interpretation is crucial to become familiar with the topic and to develop new ways of seeing the problem.

There are many ways to achieve this, and many ways of judging the quality of how a problem is framed.

INTRODUCTION



Discover

Identify & understand opportunities & needs collaboratively through co-creation with stakeholders

with **Empathy**
the ability to understand and share feelings of others, without judgments.

Define

Interpret & reframe needs & map them into activities, functions & representations

with **Mindfulness** paying attention in the present moment on purpose, without judgments.

Develop

Ideate & model concepts based on identified opportunities

with **Joy**
happiness & delight coming from a genuine sense of well-being

Deliver

Iteratively prototype & test concepts & models with users

with **Non-Attachment**
not to hold onto or grasp tightly what is considered as me or mine (e.g., my ideas, my solutions.)

Mindset

Mindfulness

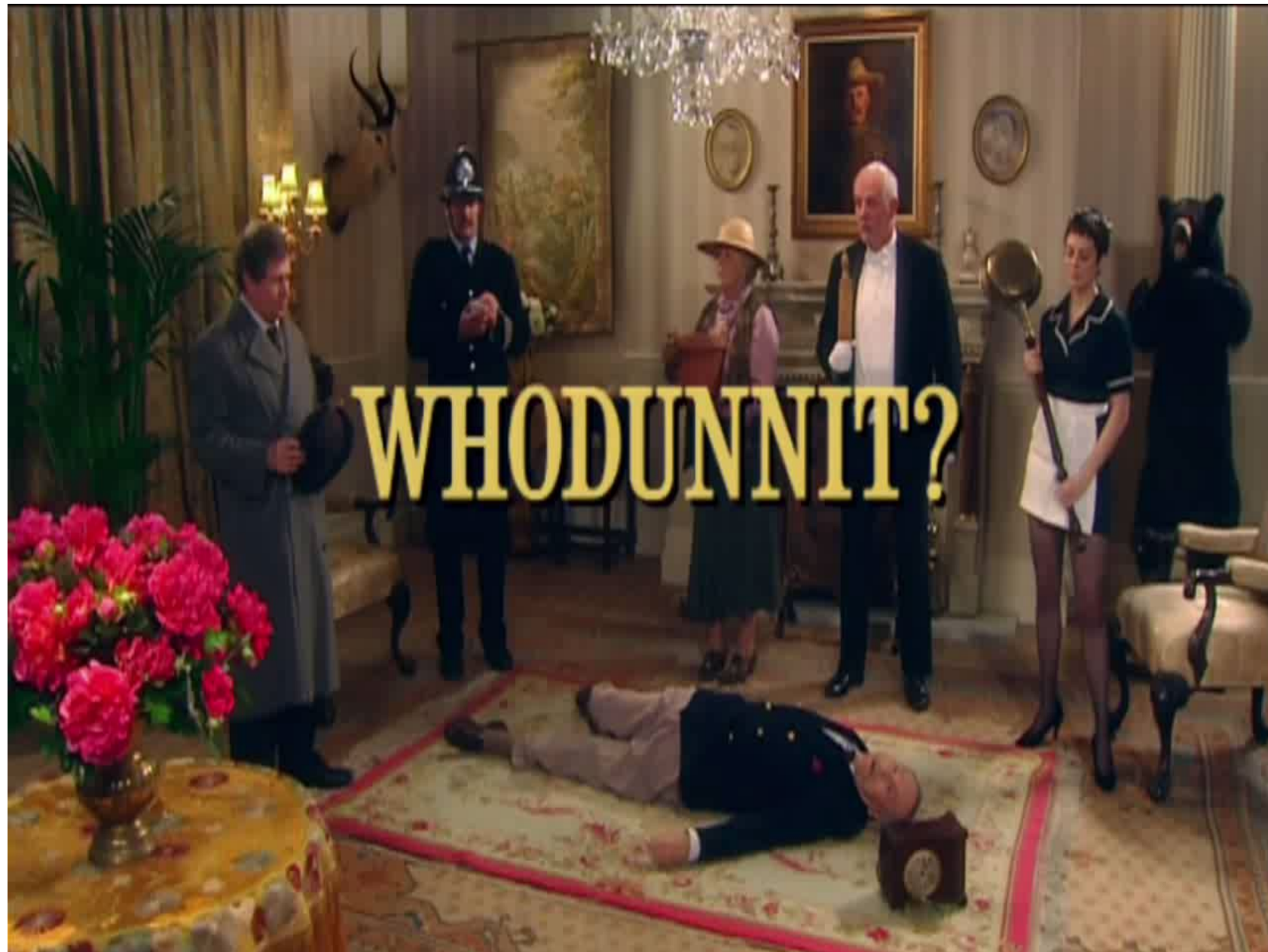




**Paying attention to the present moment,
on purpose, without judgments**

Self-Assessment

On a scale of 1-10,
how would I rate
my level of
mindfulness?



Science



OF THE TIME OUR ATTENTION IS WANDERING
(meaning, we are not paying attention to the here & now)

Killingsworth MA, Gilbert DT. A Wandering Mind Is an Unhappy Mind. *Science* 12 November 2010:
Vol. 330. no. 6006, p. 932 DOI: 10.1126/science.1192439

Science

46.9%

**OF THE TIME OUR ATTENTION IS WANDERING
(meaning, we are not paying attention to the here & now)**

Killingsworth MA, Gilbert DT. A Wandering Mind Is an Unhappy Mind. *Science* 12 November 2010:
Vol. 330. no. 6006, p. 932 DOI: 10.1126/science.1192439

What Can You Be Mindful (Or Aware, Here and Now) Of?

Seeing...
sights

Hearing ...
sounds

Tasting...
tastes

Smelling ...
scents

Feeling ...
sensations

Your body
posture

Your/ Other
thoughts

Your/ Other
emotions

The condition
of your mind



**Paying attention to the present moment,
on purpose, without judgments**

Mindfulness is a superpower to help us to come up with good insights and foresights.

Mindfulness builds upon Empathy and Listening

Empathy + Listening + Mindfulness

=

**Superpower to develop Insights or
Foresights which then help you
Define your Problem statement
(How-Might-We...)**

Method

Writing Insights



What is an insight?

A good insight has
creative tension in the statement,
and **evokes an “A-ha”**.

Structure #1

Of a typical insight



Example

Of an insight

80% of millennial youths in Asia know they are not properly saving for retirement **but** they are unwilling to change their lifestyle to save more.

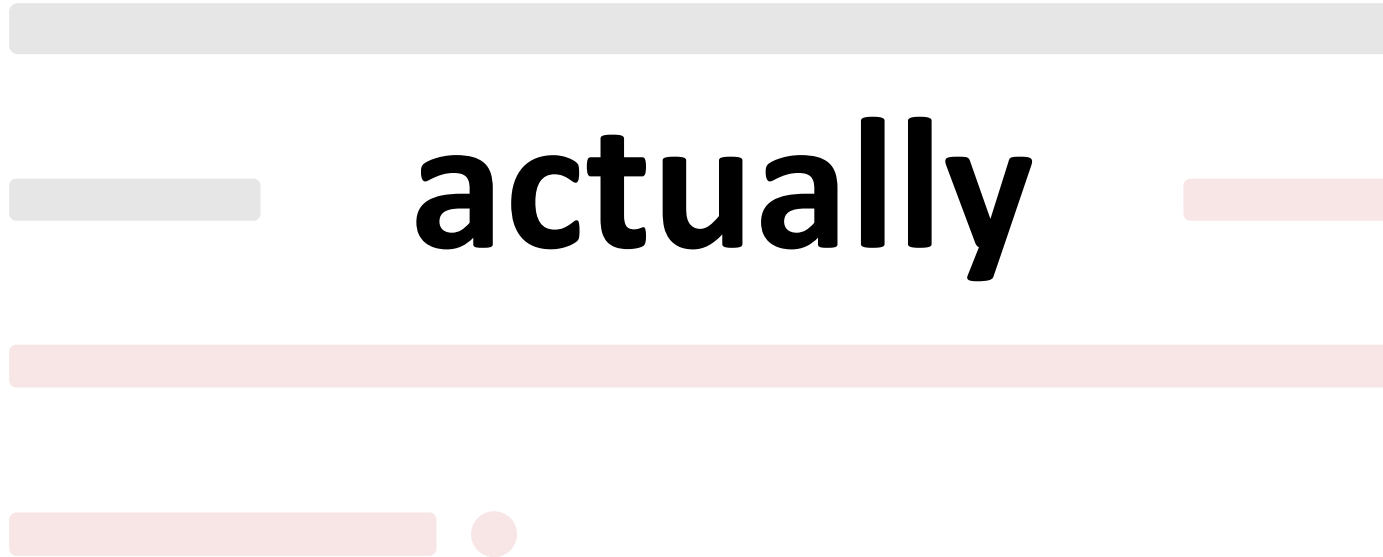
Example

Of an insight

Secondary 2 students love to play soccer because they get to make friends **but** sometimes, they **actually** make enemies instead when they play rough to win.

Structure #2

Of a typical insight



Example

Of an insight

Many people **actually** don't know
if they have insurance.

Example

Of an insight

Many people who are elderly are **actually** very comfortable to talk about dying. It's their kids who are afraid.

Example

Of an insight

Many university students **actually** skip classes where they find that they can google the information online.

Discovering Our Opportunity

How Might We Statements



With every problem,
comes an opportunity to
solve something.

How Might We statements are a way for us to *reframe* those problems into opportunities.

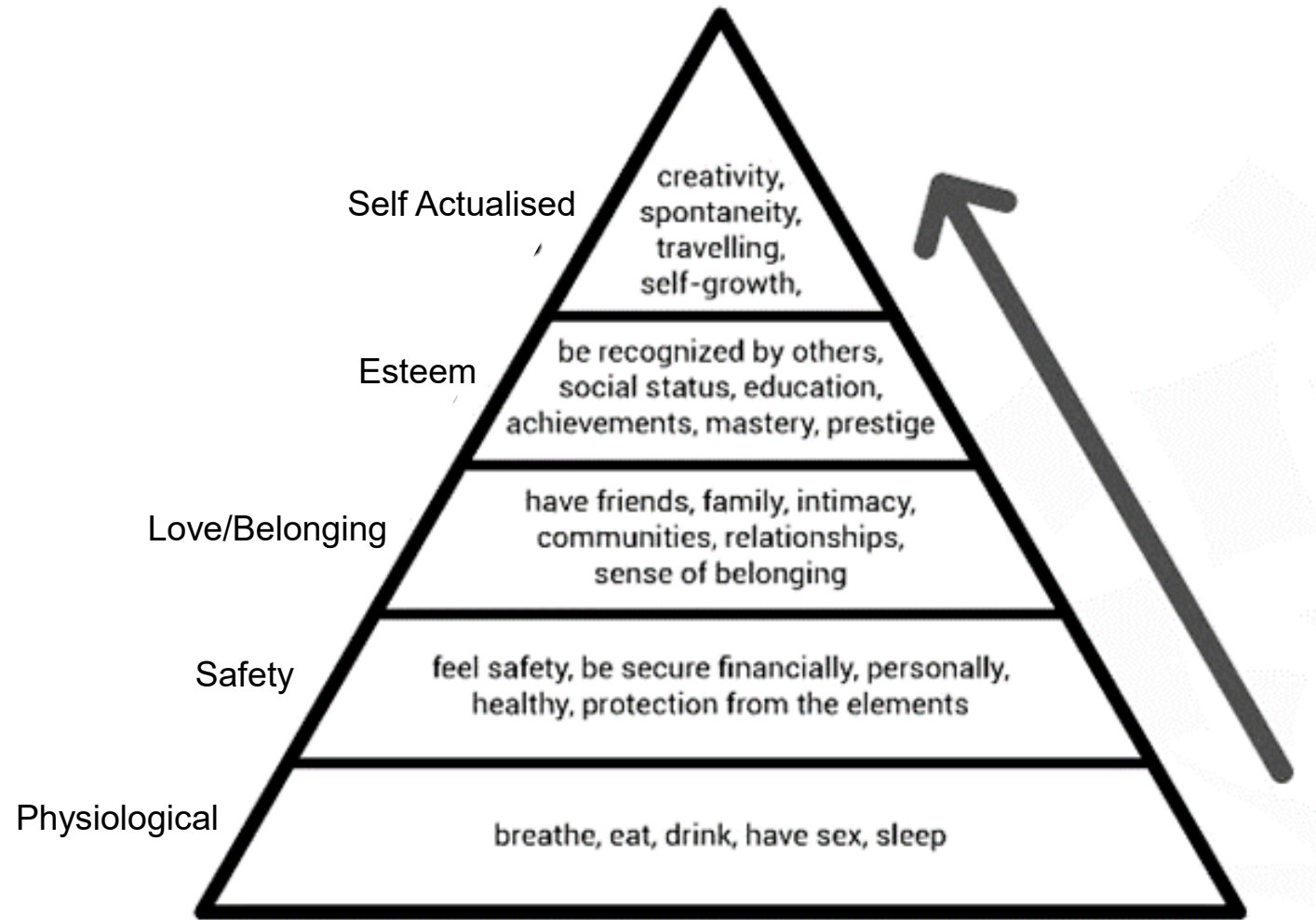
Key Terms

Users

People that seek to use our Product, Service or System (PSS)

User Needs

What users desire in the PSS



Poorly crafted opportunity statements

 *“I want to help people lower their energy consumption by automating lights at home”*
(Jumps straight to a specific solution)

 *“Buildings should promote courteous behaviours between dwellers”*
(Based on personal biased impressions, easy to challenge if no further baseline data or concrete examples are given)

 *“Students need a system to locate their lecturers outside class hours”*
(Fails to inspect actual problems and lacks a basic understanding of the teaching/learning dynamics. How can student's queries be answered more efficiently and effectively? How can they learn better?)

 *“We will design an electric baby stroller”*
(Why? Who needs it?)

How do we craft a HMW Statement?



What are we trying to achieve?



Whom are we designing for?



How do we define success in solving the problem? How do we measure?



What's the context, situation, scenarios, environment?

Example 1: World class library

The Problem/Opportunity Space

“We’ve got approvals to build a new library. We don’t want it to be just *any* library.”



Example 1: World class library

The Problem/Opportunity Space

“We’ve got approvals to build a new library. We don’t want it to be just *any* library.”

Reframed into...

The Opportunity Statement

HMW **design a new local library** so
that it **meets world-class standards** *What?*
for **both visitors and staff?** *What does success look like?*
For whom?



Example 1: World class library

What?

HMW design a new local library so
that it meets world-class standards *What does
success look like?*
for both visitors and staff? *For whom?*

Stakeholders could include:

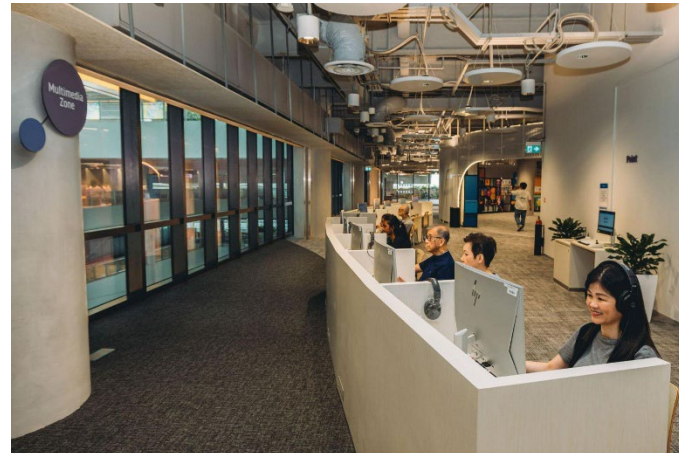
- Young families who live in the area
- Elderly folk who want to hang out somewhere quiet
- Students who want somewhere to study during their time off
- Wheelchair-bound users
- Special needs users
- Young children
- Librarians
- Cleaning staff
- Programming staff

Example 1: World class library

What?

HMW design a new local library so that it meets world-class standards *What does success look like?* for both visitors and staff? *For whom?*

Along with beautiful new communal facilities, the new Punggol Regional Library officially opened in 2023 has a focus on enabling persons with special needs to be self-reliant in the library as well as being inviting to young families.



Example 2: End of life conversations

The Problem/Opportunity Space

“I don’t feel comfortable talking to my elderly parents about what happens after they pass on. I think it’ll be really awkward and seen as disrespectful even though it’s very real.”



Example 2: End of life conversations

The Opportunity Statement

What?

HMW reduce taboo around
end-of-life for elderly and their
loved ones so they feel ease
discussing the emotional and
practical concerns of passing
on?

*For whom?**What does success look like?*

Example 2: End of life conversations

What?

HMW reduce taboo around end-of-life for elderly and their loved ones so they feel ease discussing the emotional and practical concerns of passing on?

*For whom?**What does success look like?*

Stakeholders could include:

- 75-90 year olds
- 60-75 year olds
- Immediate family
- Extended family
- Close friends
- Bereavement counselors
- Community agencies
- Policy makers
- Hospice associations
- Hospice staff

Example 2: End of life conversations

What?

HMW reduce taboo around end-of-life for elderly and their loved ones so they feel ease discussing the emotional and practical concerns of passing on?

What does success look like?

The HappyUrns project included a Celebration Kit of prompts surrounding memories and personal stories, Life in a Year book to create a journey of a year in your life for a loved one to experience, and the HappyUrns virtual exhibition showing off beautiful reimagined urns and their owners who co-created on them.

Created in partnership with Lien Foundation.



Theresa's Urn

Witty, sharp, and chuddlesome, Theresa was a Bible studies teacher for five decades. She enjoys doing crafts in her free time and creating Chinese ink-inspired paintings. A deep lover of Chinese tradition, she chose an immortal peach as inspiration for her urn and made it her favorite color – a beautiful pale jade. Theresa's urn has a special compartment she made to store a necklace she treasures. During the design process, Theresa would often laughingly comment on designs saying the ideal case would be for "S.B." which means more greenery in the design.

Jade Peach Jar

Pale Jade Colour
Immortal Peach
Necklace Holder
Roses and "S.B." (grass)



Rainbow Pebble

Rainbow
Pebble Shape
Tactile Swirl Pattern

Mary's Urn

A spirited character and a music teacher in her younger years, Mary's love of life is something that carries through her interactions with everyone she meets. Being bed-bound doesn't stop her jovial attitude in the least. When asked if she could have any colour for her urn design, she asked: "would it be possible to have a rainbow?" She picked out a pebble-inspired shape with a wavy pattern. With her ailing vision, care was taken for the surface of the urn to be treated to have texture and grooves so that she could feel it as well.



Nancy's Urn

Nancy loves precious objects, from gold to porcelain. You can often find her at the home decked out in some of her newest finds. It could be necklaces, bracelets, or shiny gold nail polish. She's a dear who loves to show you what her latest colour choice for nail polish is, be it shiny gold or bold red. Just like the playful monkeys that are her favourite animal, Nancy is likewise playfully insistent that the best shape for an urn is a sphere. Her urn design is a blue and white porcelain inspired sphere, with floral motifs, reflecting her enjoyment of nature. If you ever meet her at the nursing home, she'd be very happy to show you her latest choice of nail colours, and discuss with you on how curry chicken is the best tasting food.

Porcelain Orb

Floral porcelain motif
Spherical
Luxurious Design



Example 3: Public spaces

What?

For whom?

HMW encourage Toronto residents to spend more time outdoors together, so as to activate public spaces?

What does success look like?

Example 3: Public spaces

What?

For whom?

HMW encourage Toronto residents to spend more time outdoors together, so as to activate public spaces?

What does success look like?

Stakeholders could include:

- Elderly Residents
- Families
- Young working professionals
- Commuters
- Developers
- Business Owners/ shopkeepers
- Commercial Tenants

Example 3: Public spaces

What?

For whom?

HMW encourage **Toronto residents** to spend more time outdoors together, so as to **activate public spaces?**

What does success look like?

Diverse program mixes and flexible space use leading to theoretically higher usage and activation of public space.



Ground-floor space that is activated 33% more time each day

A typical street in Quayside would have a more diverse program mix and more flexible co-tenancy options, leading to three hours more daily activity relative to the weighted average of Toronto street activity today.

9hrs

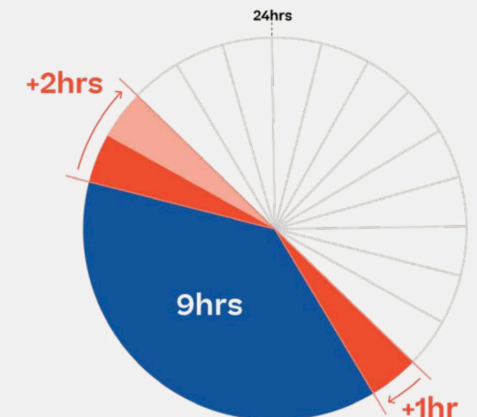
10am - 7pm

Average street activity today across destination, local, and downtown streets

12hrs

9am - 9pm

Average street activity for the Sidewalk Toronto project



This same area space would be occupied by a band from 7pm to midnight.

HMW redesign the future of BE?

Examples of built environment HMW statements

#LowCarbon

HMW make low-carbon design highly desirable at scale for industry stakeholders, so as to reduce the high carbon footprint construction projects have on the environment?

#Supercollaboration

HMW improve collaboration, design, and construction processes among different stakeholders involved in projects in Singapore, in order to minimise miscommunication and enhance workflow efficiency?

#PopulationChange

HMW transform old housing estates in Singapore to accommodate changing demographics, higher population density, and promote eco-friendly, sustainable living?

#ConsiderateConstruction

HMW minimise construction disruptions and mitigate noise pollution for residents and commuters in Singapore, while ensuring efficient construction processes?

#TalentManagement

How might we increase the retention of fresh graduates in the Built Environment industry in Singapore, fostering their long-term commitment and professional growth so as to avoid a local brain drain in the industry?

#TechTransformation

HMW harness new technologies to overcome the inconvenience of tedious/repetitive tasks, so as to maximise efficiency and create capacity to tackle larger issues?



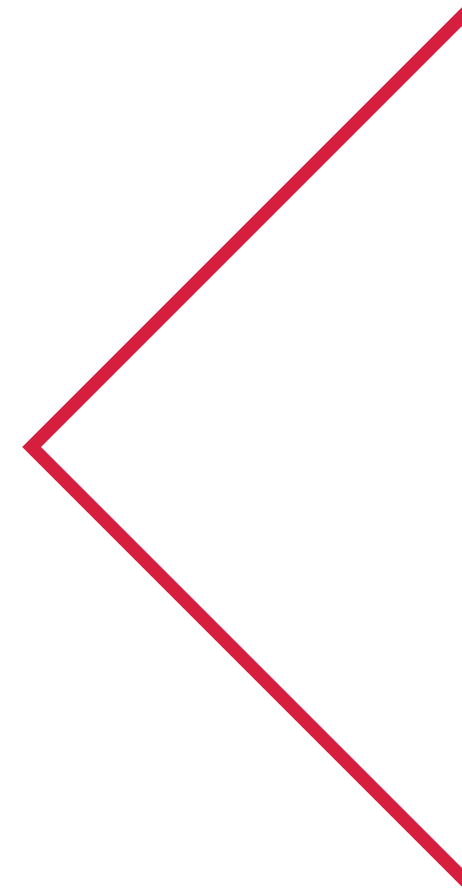


It's Your Turn!

HMW

1. Discuss with your team and craft 1 HMW Statement you would like to work on together.

- ✓ Pick something you and your group cares about.



How Might We w/ AI



Example

The Problem/Opportunity Space

“We are building a new library, but we don’t want it to be just *any* library.”

↓
Reframed into...

The Opportunity Statement

HMW **design a new local library** so
that it **meets world-class standards**
for **both visitors and staff?**

What are we creating?

What does success look like?

For whom?



Example

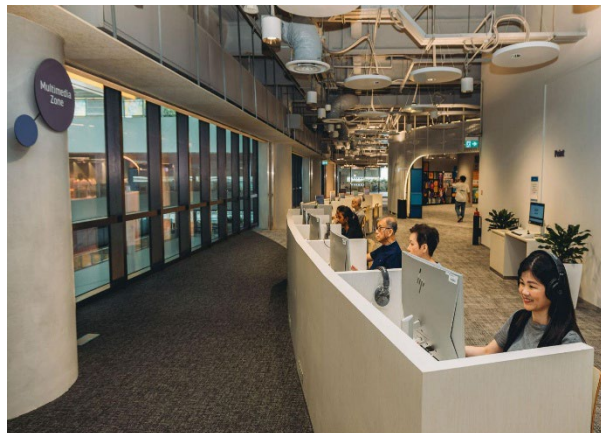
What are we creating?

HMW design a new local library so that it meets world-class standards for both visitors and staff?

What does success look like?

For whom?

Along with beautiful new communal facilities, the new Punggol Regional Library officially opened in 2023 has a focus on enabling persons with special needs to be self-reliant in the library as well as being inviting to young families.



Example

For whom?

HMW encourage **Toronto residents** to spend more time outdoors together, so as to **activate public spaces?**

What?

What does success look like?

Diverse program mixes and flexible space use leading to theoretically higher usage and activation of public space.

Ground-floor space that is activated 33% more time each day

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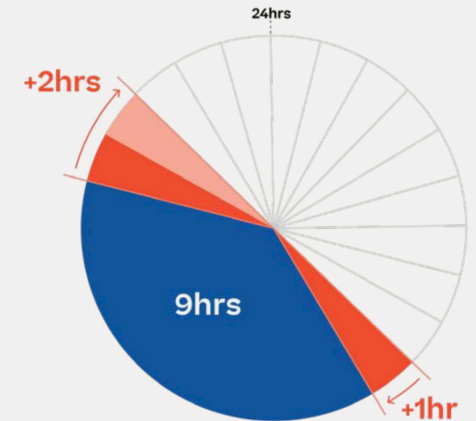
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12hrs

9am - 9pm

Average street activity for the Sidewalk Toronto project





It's Your (+AI) Turn!

HMW

1. Formulate an initial HMW
2. Gather a list of user needs and critical consideration
3. Feed the initial statement, and user needs opportunities into ChatGPT:

"Using the following initial HMW statement [...] and the accompanying list of user needs [...], create a refined HMW statement that fosters greater potential for innovative solutions within the context of a 13-week design course."

4. Iterate and refine as needed such as by asking for personas and then refining the HMW for a relevant group of users or stakeholders



Tips and Tricks

1. If we ask ChatGPT to generate a refined HMW, it will respond with something that is overly generalised.



To improve upon this...

2. Try to ask ChatGPT to generate needs or personas, then feed those needs and detailed personas into the prompt afterwards. The refined HMWs will be more meaningful and precise.

Example

HMW

1. *“HMW: develop a next generation EV” (initial prompt)*

NEED

2. *“What are some potential user needs for an EV?”*



ChatGPT Output

1. Performance

Extended range (400+ miles).

Fast charging (10-20 minutes for 80%).

Enhanced energy efficiency and regenerative braking.

2. Cost & Sustainability

Affordable pricing and maintenance.

Long-lasting, eco-friendly batteries (10+ years, recyclable materials).

Lower lifecycle emissions.

3. Charging & Infrastructure

Wide, reliable charging networks.

Smart and wireless charging options.

Vehicle-to-grid (V2G) energy integration.

3. Refined HMW

What?

“HMW design **electric vehicle systems** that seamlessly integrate high performance, sustainability, and user-centric features to re-define the **driving experience for urban citizens in diverse and challenging environments?**”

What does success look like?

For whom?

Why this HMW works:

- 1.Focuses on integration:** Encourages students to think holistically about combining performance, sustainability, and user experience.
- 2.Allows broad exploration:** "Redefine the driving experience" leaves room for creative interpretation and a range of innovative ideas, from user interfaces to modular systems.
- 3.Challenges conventional thinking:** "Diverse and challenging environments" inspires solutions that go beyond standard urban or highway settings, potentially sparking ideas for adaptability and inclusivity.
- 4.Feasible within 13 weeks:** Students can prototype specific subsystems or conceptualize bold solutions while grounded in real-world constraints.
If you'd like a more specific focus (e.g., battery systems, charging, or connectivity), I can refine it further.

Example

HMW

1. *“HMW: develop a next generation EV” (initial prompt)*

NEED

2. *“What are some potential personas for an EV?”*



ChatGPT Output

1. Alex, City Professional (28)
2. Maria, Suburban Caregiver (35)
3. Priya, Tech Enthusiast (30)
4. Kevin, Value Seeker (45)
5. Sarah, Logistics Coordinator (50)
6. Jake, Outdoor Enthusiast (40)
7. Victor, Business Executive (55)
8. Liam, Emergency Responder (38)

4. Further refined HMW for one of the personas “Emergency Responder”

For Persona 8: Emergency Responder (Liam, 38) here’s a redesigned "How Might We" (HMW) tailored to their unique needs and challenges:

What?

What does success look like?

"How might we design high-performance electric vehicles that ensure reliability, rapid response, and operational efficiency for emergency responders in extreme and unpredictable conditions?"

For whom?

Why this HMW works:

- 1.Focus on emergency needs:** Highlights the critical role of speed, reliability, and efficiency in emergency scenarios.
- 2.Challenges conventional EV design:** Encourages exploration of robust solutions for harsh climates, terrains, and unpredictable situations.
- 3.Room for innovation:** Opens possibilities for features like mobile energy storage, rapid charging, and specialized equipment integration.
- 4.Emphasizes practical use:** Grounded in operational realities like uptime, durability, and rapid deployment.

Real and not Fictitious are best

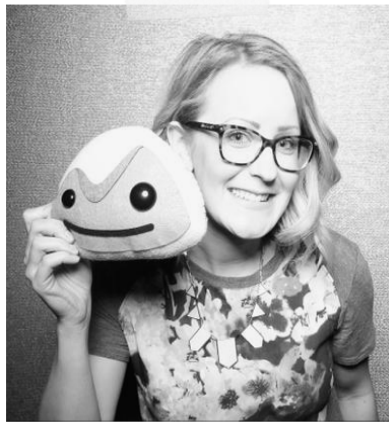
Detailed Personas



Persona: Example

MYRA TRAINOR

INVOLVED AND INVESTED



AGE: 42

RELATIONSHIP: Single

OCCUPATION: Creative Director

PROPERTY: One, short-term

LOCATION: Coogee

LANGUAGES: English

DESCRIPTION

Myra self-manages her studio apartment and rents it out occasionally through Airbnb. She has trouble managing her keys during guest transitions and would always depends on her friends when she's not around.

GOALS

- I want my guests to have a great experience.
- Improve my property rating and online presence.
- Rent out my unit for more money.

"I get excited when my guests are happy. That motivates me to self-manage my property but managing the transition gives me headaches."

.... TECHNOLOGY



..... BELIEFS

FRUSTRATION

AGENTS, TENANTS, AND SERVICES

TRUST

AGENTS, TENANTS, AND SERVICES

INVOLVEMENT

AGENTS, TENANTS, AND SERVICES

BEHAVIOR AND HABITS

- I take time to curate and assess quality cleaners during empty periods.
- I do most of the cleaning myself when there are quick transitions.
- I always think about ways to manage key transfers better as I frequently travel.
- During my free, time I look at other online providers like Homeaway and VRBO for listing my property to get more rentals.
- I spend a great deal of time & effort managing my property as my online ratings are important.

FRUSTRATIONS

- I have trouble managing my keys during guest transitions.
- I am security cautious about who handles my keys, and usually only trust friends.
- I need to be available to respond to guests.
- I have high interactions with guests which sometimes is very stressful as most enquiries, emergencies, and general work comes to me.

FEATURES

- Manage guest transitions.
- See reviews/ratings on cleaning services.
- Provide a welcome pack containing ground rules, instructions and a list of my emergency contacts available to them.
- A way to better communicate and build trusts with my guests.

Persona: Example

BRAD THOMPSON

ALL STAR AGENT



AGE: 35

RELATIONSHIP: Married

OCCUPATION: Real-estate Agent

PROPERTY: 150 long-term residential (managing)

LOCATION: Paramatta

LANGUAGES: English

DESCRIPTION

Brad has managed 50 residential properties for 5 years. He works in the real-estate industry and likes getting involved but would prefer to automate regular tasks between his tenants. This would allow him to manage more properties whilst maintaining his high quality of service and upholding his all-star reputation.

"I do everything that my owners don't want to do. My role is to make their life easier."

.... **TECHNOLOGY**



..... **BELIEFS**

FRUSTRATION

OWNERS, TENANTS, AND SERVICES

TRUST

OWNER, TENANTS, AND SERVICES

INVOLVEMENT

OWNERS, TENANTS, AND SERVICES

BEHAVIOUR AND HABITS

- Landlords often trust me with their properties because I am recommended through word-of-mouth.
- I'm always looking for another property to manage.
- I often manage on-going rental issues like late rent payments, repairs, complaints, etc.
- I have to organise and manage Strata, builders, inspections, and 3rd-party contractors.
- I often need to assess the quality of service from 3rd-party contractors.

GOALS

- Increase my property portfolio.
- As much as I like to be involved in managing my properties, I wish regular tasks like tracking rental payments or organising repairs were automated.

FRUSTRATIONS

- Owners have very high expectations and the volume of requests can be hard to manage. Sometimes conflict arises due to lack of knowledge in the property field.
- When things pop up and I have to get involved, it means I can't take time off after hours.

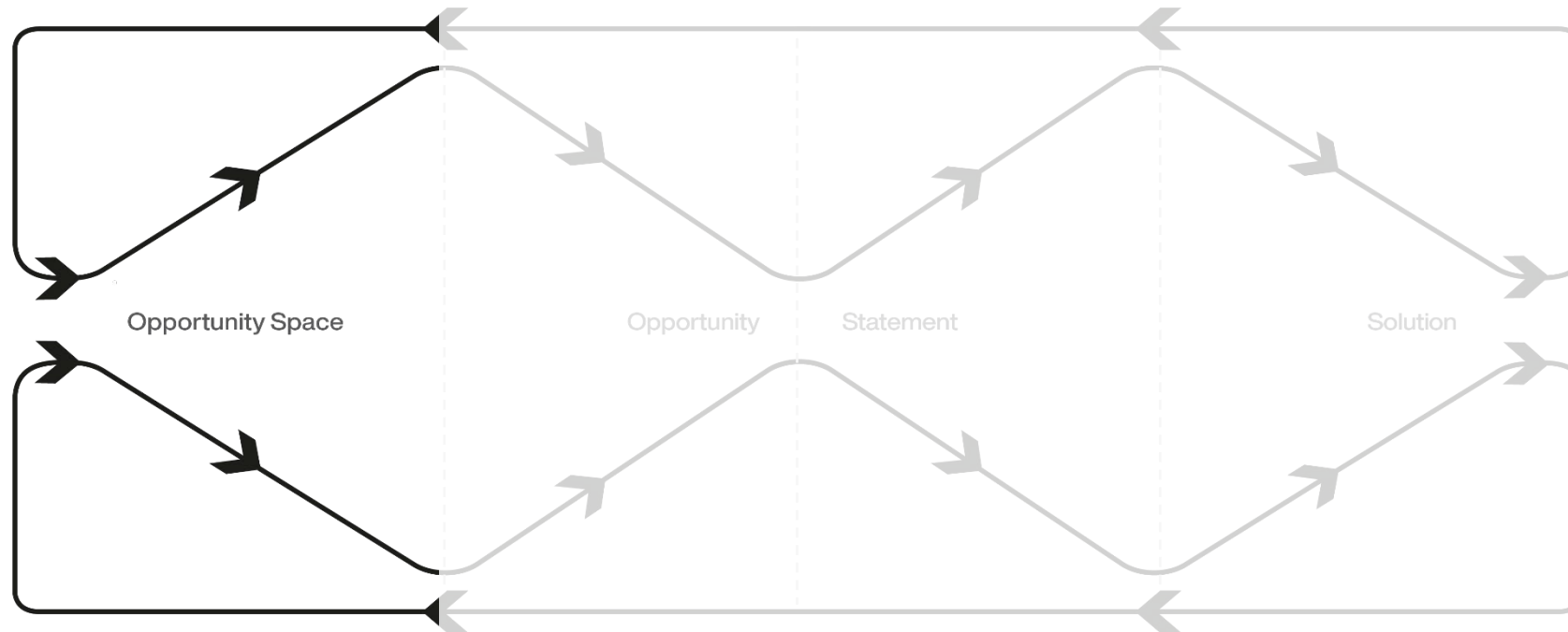
FEATURES

- I wish I had a way to monitor my tenants.
- The ability to see reviews of 3rd-party contractors.

Discovering Our Opportunity

User Journey Map





Discover

Identify & understand opportunities & needs collaboratively through co-creation with stakeholders

with **Empathy**
the ability to understand and share feelings of others, without judgments.

Define

Interpret & reframe needs & map them into activities, functions & representations

with **Mindfulness** paying attention in the present moment on purpose, without judgments.

Develop

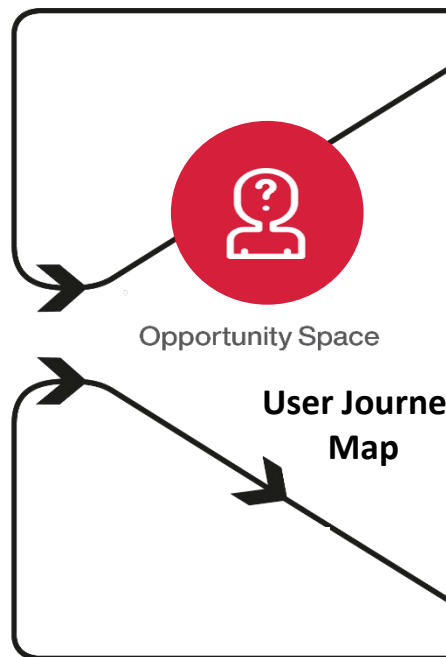
Ideate & model concepts based on identified opportunities

with **Joy**
happiness & delight coming from a genuine sense of well-being

Deliver

Iteratively prototype & test concepts & models with users

with **Non-Attachment**
not to hold onto or grasp tightly what is considered as me or mine (e.g., my ideas, my solutions.)



Discover

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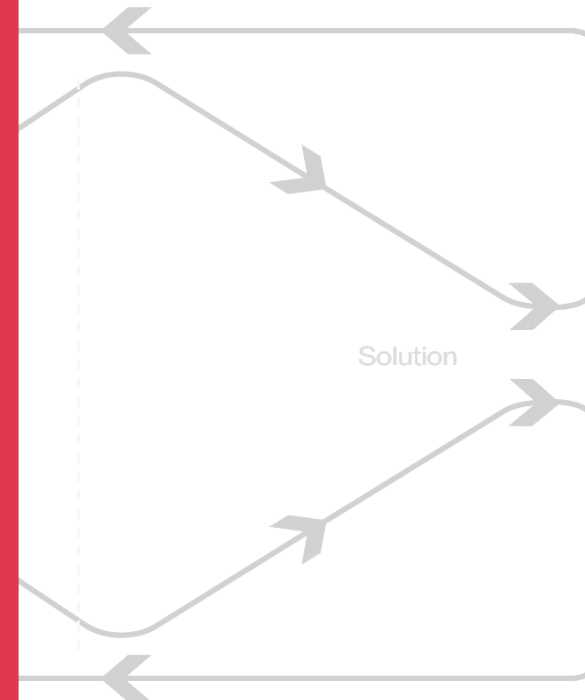
Identify and understand opportunities and needs collaboratively through co-creation with stakeholders

Understand

- Who are the users and stakeholders?
- What are their needs?
- How might we delight them and their experiences?
- What are their actions, reactions and emotions?
- What is the context?
- What research and user studies are needed?

Empathise

- How do users behave?
- How do they feel?
- How do we see through their eyes?
- What extreme conditions may inform us?
- How do they interact with objects, the environment, and each other?



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User Journey Map

Gives a *visual or graphic interpretation* of the overall story from an *individual's perspective* of their relationship with the organization over time and across channels.

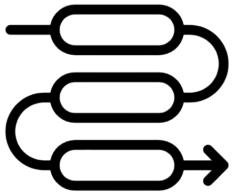


Told from the user's perspective but also emphasizes the important intersections between user expectations.

User Journey Map: Objective



Develop empathy by understanding experiences.



Uncover insights through detailed process analysis.



Identify user journey touchpoints and milestones.

User Journey Map: Process

1. *Choose a persona and a scenario*

Clarify the persona's needs, expectations and goals within the defined scenario.

2. *Map the journey*

Plot the relevant plots of action between user and the Product, Service or System (PSS)

3. *Identify gaps and insights*

Analyse the Journey Map and identify interactions that are pain points and areas where user experiences delight. Draw **insights** to improve on the user experience.

Vocabulary

Touchpoints

User actions and interactions with the organization. This is what the user is doing.

Channels

Where interaction takes place and the context of use. A channel defines the opportunities or constraints of a touchpoint.

Emotional Response

Peaks and valleys illustrating frustration, anxiety, happiness etc.



Personas



Emotional
Response



Timeline



Touchpoints
and Channels



Scenarios

Steps to Make a User Journey Map

1. Review Goals

Consider organizational goals and specific goals

2. Gather Research

- User/ Customer Interviews
- Ethnography & contextual inquiry
- User/ Customer surveys
- User/ Customer support or complaint logs
- Web analytics
- Social media listening
- Competitive intelligence.

Steps to Make a User Journey Map (cont.)

3. Ideate and Identify Touchpoints and Channels

As a team, generate **a list of the user touchpoints** (actions and interactions with the organization) and the channels on which those touchpoints occur today.

4. Create Journey Map

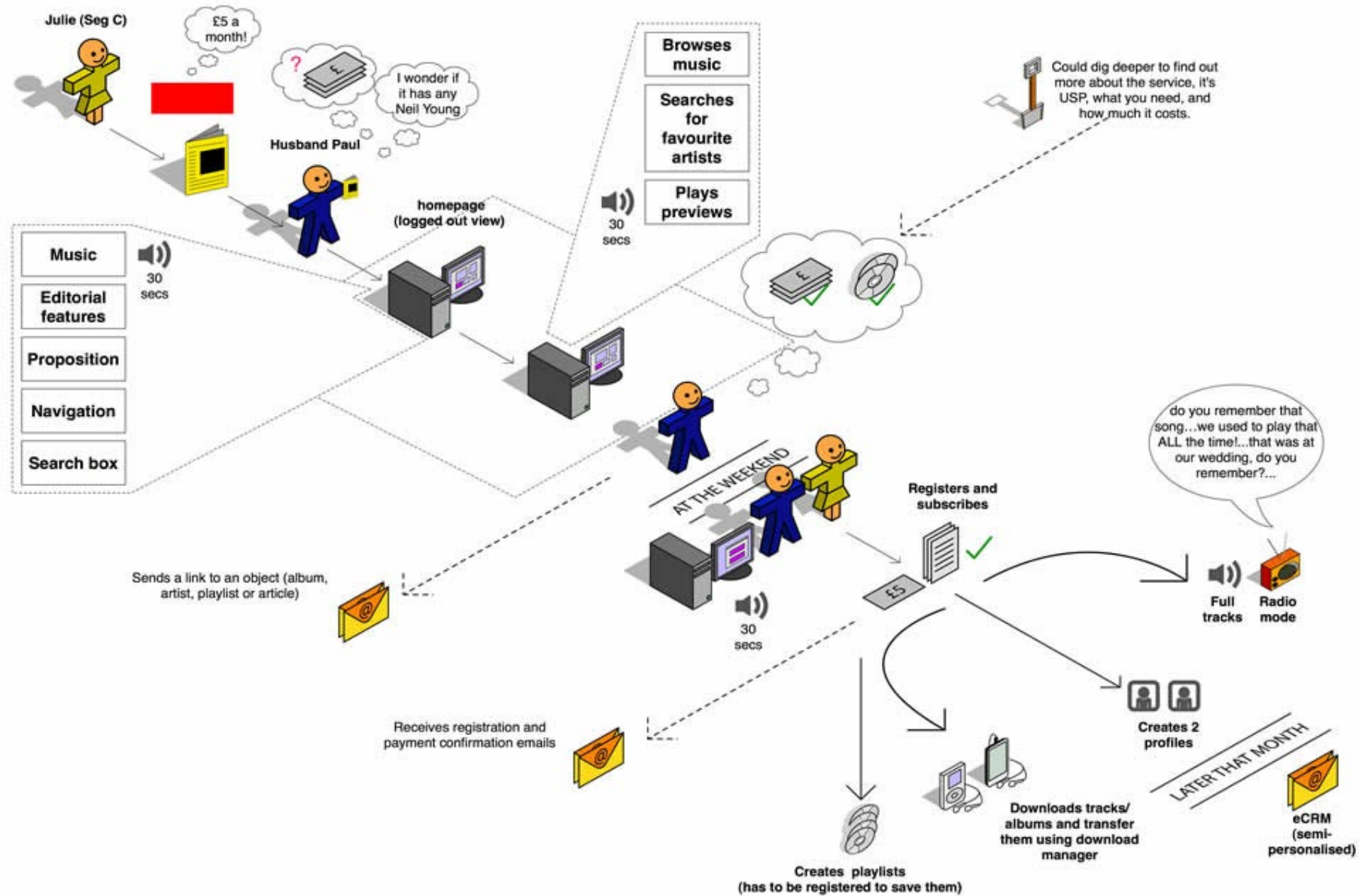
The different aspects of a persona and their experiences in a certain situation are shown in journey maps. The objective is to gain a comprehensive understanding of what it's like to be that persona during this event, with an emphasis on their thoughts, feelings, sight, sound, words, and actions.

5. Refine and Digitise

6. Share and Use (and Iterate with feedback!)



Example: Music Purchasing



Example: Shopping for a Car

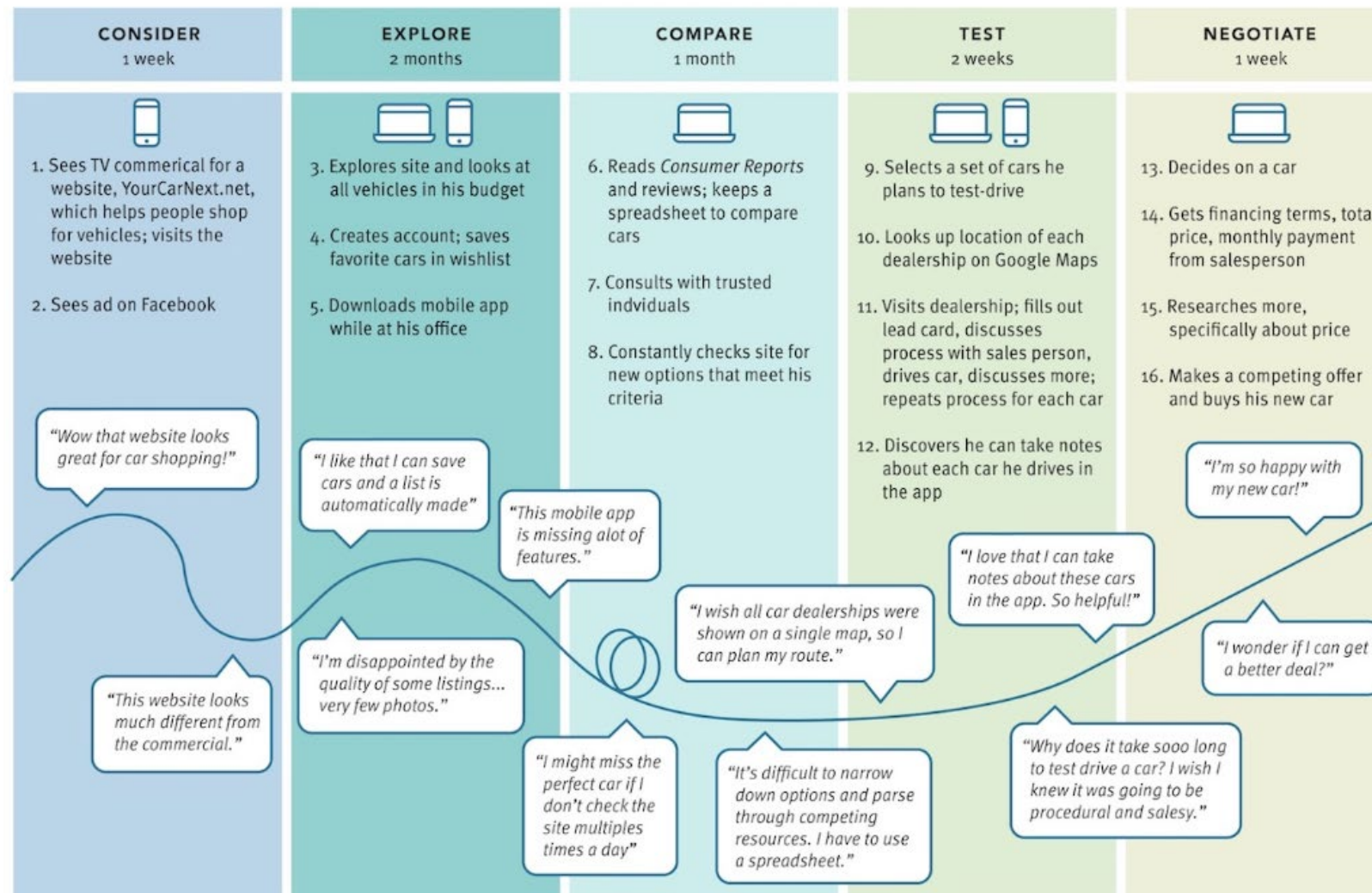


EMOTIONAL ERIC

Eric is an emotional car buyer. He purchases based on aesthetics and status.
Scenario: Eric recently moved to the area. He is shopping for a car that is fun to drive and dependable enough for use for everyday commuting.

EXPECTATIONS

- Ability to compare cars and their breakdowns
- Good photography with closeups, inside and out
- Video overview of car with demonstrations



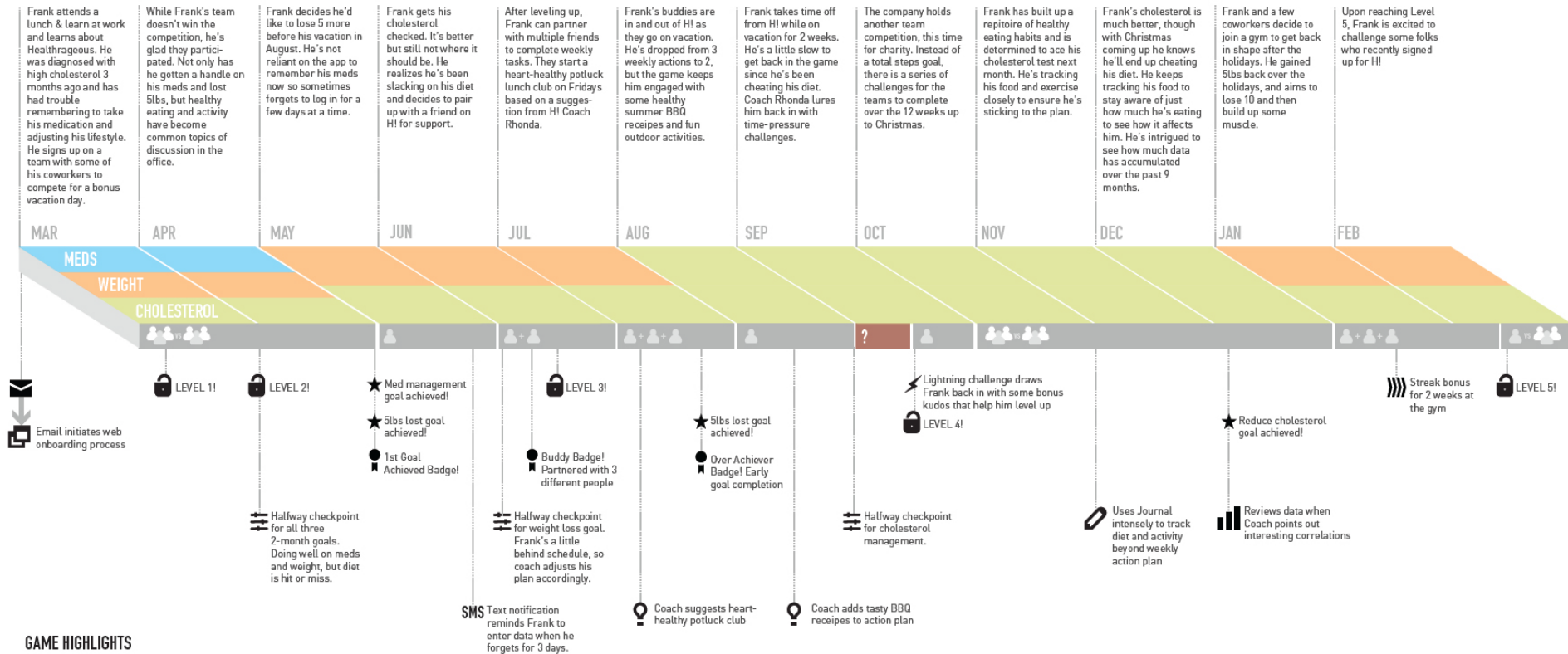
Example: Patient Action

FRANK

43, Call Center Manager, Family Man
Recently diagnosed with high cholesterol

GADGETS: Pedometer, Wireless Scale in Office, Smartphone

GOALS: 100% Med Adherence, Lose Weight, Reduce Cholesterol



GAME HIGHLIGHTS

SCHEDULED NOTIFICATIONS

DAILY Mobile medication reminders & logging

FRIDAY End-of-week checkpoint

SUNDAY Summary of past week

MONTHLY Goal progress & maintenance report

SAMPLE ACTION PLANS & KUDO SCORES

WEEK 1 [MAR]

- ✓ Take medication when reminded by HI mobile app
- ✗ Walk 2500 steps per day [for team competition]
- ✓ Buy olive oil to use instead of butter

2 1 per action

WEEK 8 [APR]

- ✓ Take medication without being reminded by HI mobile app
- ✓ Walk 4500 steps per day [for team competition]
- ✓ Play frisbee with the kids

18 1 per action + x2 completion bonus + x2 competition kudos for 3rd place

WEEK 18 [JUL]

- ✗ Cook 3 healthy meals this week
- ✓ Go for a 5-mile bike ride
- ✓ Healthy potluck lunch [with Jim, Sally, Mark]

19 1 per action + 8 co-op bonus + 10 goal completion bonus

WEEK 22 [AUG]

- ✓ Healthy BBQ with family
- ✗ Swim for 5 hours

5 1 per action - 1 sponsor penalty + 5 goal checkpoint evaluation

WEEK 27 [SEP]

- ✓ Make a healthy fruit dessert [do it tonight for bonus kudos!]
- ✓ Walk 5000 steps per day

14 1 per action + 5 lightning bonus + x2 completion bonus

WEEK 36 [NOV]

- ✓ Track meals in journal
- ✓ Eat 15 servings of veggies [for team competition]

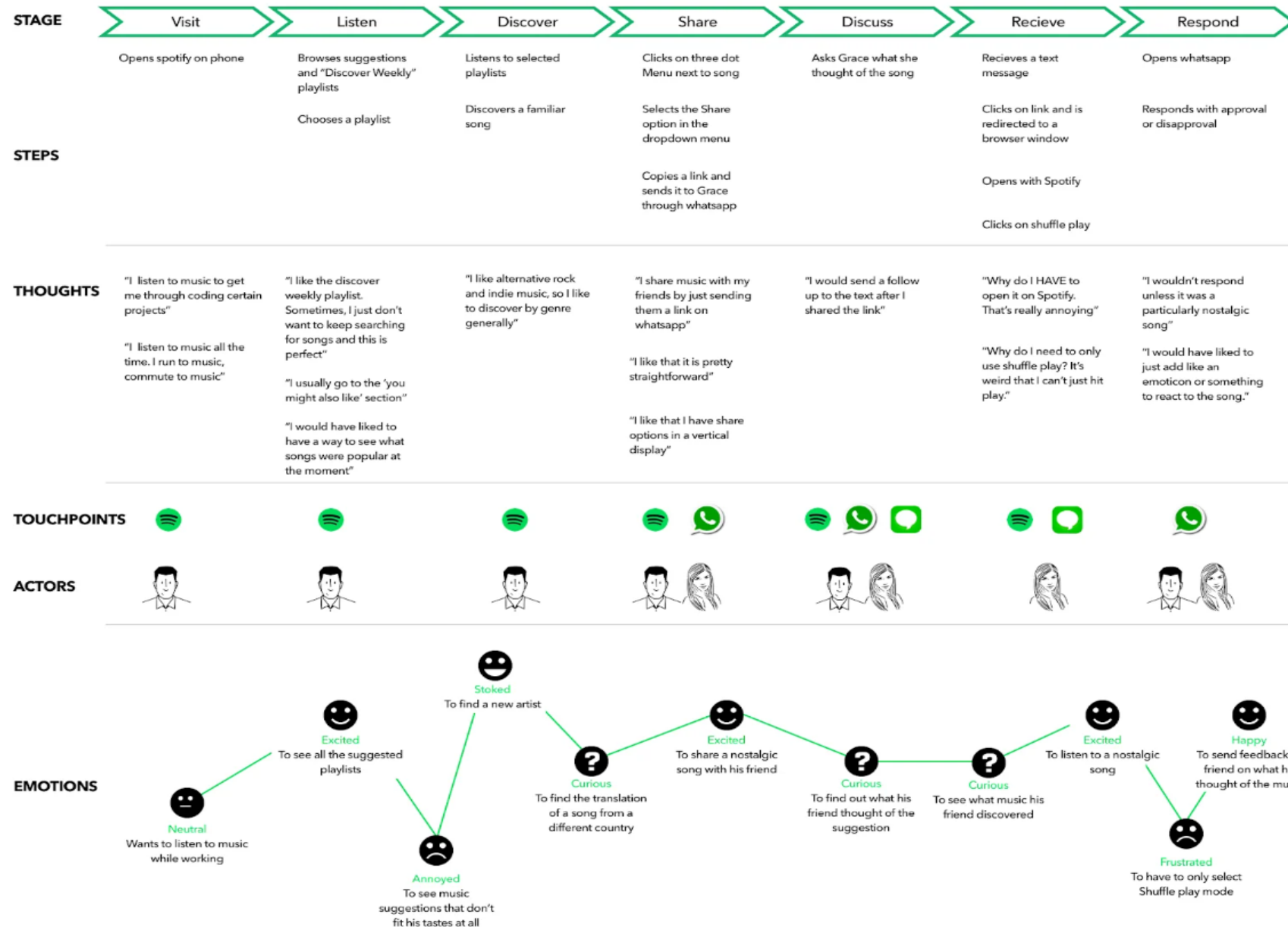
4 1 per action + 2x completion bonus

WEEK 47 [FEB]

- ✓ Go to the gym 4x
- ✓ Pack a new lunch food for you and the kids
- ✓ Walk more than a Level 1 group [versus Erik, Yang, Cory]

26 1 per action + 8 streak bonus + 4 challenge win + x2 completion bonus

Example: Spotify (i.e. music sharing experience)



Example: Airport Experience

Pre-boarding experience



Anna

ANNA'S JOURNEY

Anna checks in

Anna arrives at the airport

Anna checks out the
flight gates

Anna goes through security



Anna

ANNA'S STORYBOARD

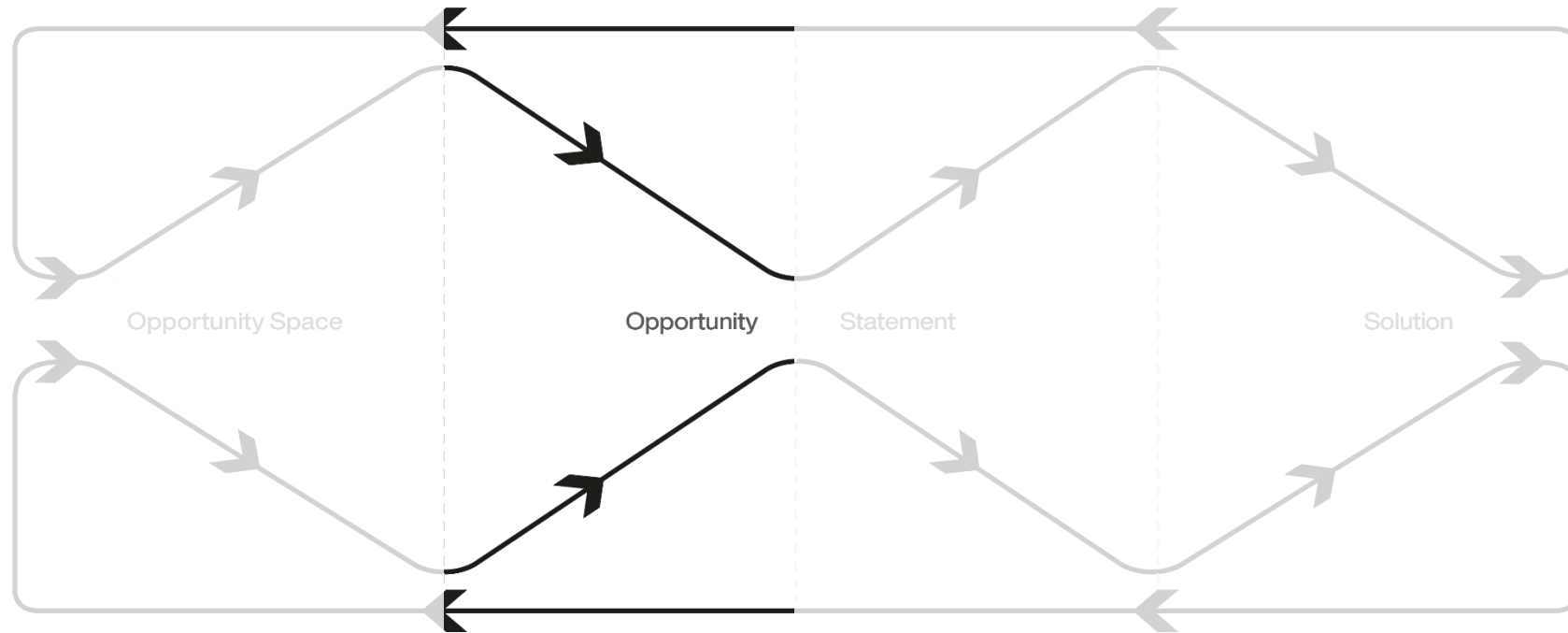


Across disciplines, problems are identified, defined and initiated differently.

In some areas, the process is viewed as '**messy**' and undetermined, others follow more systematic approaches.

In all cases, information gathering and interpretation is crucial to become familiar with the topic and to develop new ways of seeing the problem.

There are many ways to achieve this, and many ways of judging the quality of how a problem is framed.



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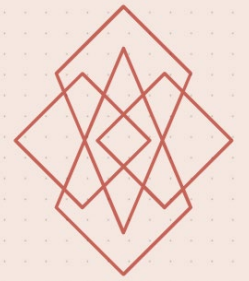
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MTD Core Module
Term 1 – Sept 2026
12 Credits – 14 Weeks



AI-Accelerated Ventures,
From Prototype to Market

01.505

Innovation by Design (IbD)

Studio – Week 2 – LC01

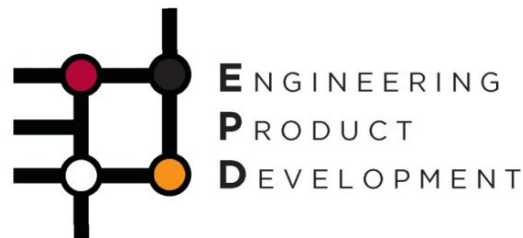
What is Frugal Innovation?

DISCOVER Phase of Design Thinking

Assoc. Prof. NAGARAJAN RAGHAVAN

1.602-35

Engineering Product Development (EPD) Pillar
Singapore University of Technology & Design



What is Frugal Innovation?

- FRUGAL INNOVATION is an engineering and innovation approach focused on creating **high-value, robust, affordable solutions** while **deliberately minimizing resources, complexity, cost, energy, and material use**.
- *Is it a new discipline?* **Not exactly—but it is emerging as a recognizable engineering field.** The underlying philosophy is old: engineers have always designed under constraints. What is newer is treating **resource scarcity and affordability as first-class design objectives**, rather than merely constraints encountered after a conventional design has been developed.

$$\text{Frugal Engineering} = \frac{\text{Useful Function} + \text{Accessibility} + \text{Reliability}}{\text{Cost} + \text{Materials} + \text{Energy} + \text{Complexity} + \text{Resources}}$$

- The objective isn't simply "**make it cheap.**" A poorly made inexpensive product isn't necessarily frugal engineering. Good frugal engineering asks engineers to maximize **value delivered per unit of resource consumed**.

Principles of Frugal Innovation

- **Essential functionality** — eliminate features that don't create meaningful user value.
- **Resource efficiency** — minimize materials, energy, computing, water, manufacturing steps, etc.
- **Affordability** — design around what users can realistically afford.
- **Robustness** — tolerate harsh environments, imperfect infrastructure and variable operating conditions.
- **Simplicity** — fewer components and simpler manufacturing, maintenance and repair.
- **Local manufacturability** — use locally accessible materials, processes and skills where appropriate.
- **Scalability and accessibility** — make the solution deployable to much larger populations.
- **Sustainability/circularity** — repair, reuse and reduced resource consumption increasingly fit naturally into the philosophy.

NOTE: Frugal Does NOT Necessarily Mean Low-Tech

An important point is that **frugal doesn't necessarily mean low-tech**. You can have:

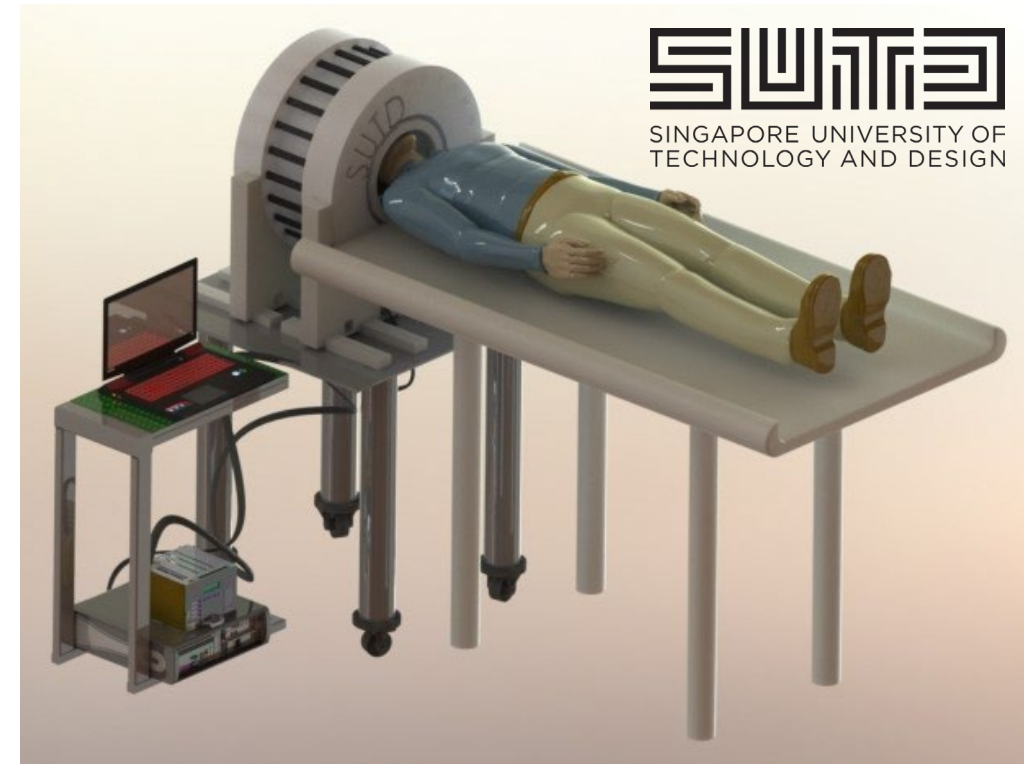
- **Frugal AI** — small models, edge inference, compressed networks
- **Frugal robotics** — inexpensive sensors/actuators and minimal hardware
- **Frugal healthcare engineering** — portable diagnostics and low-cost imaging
- **Frugal semiconductor systems** — older technology node chips are used as they are sufficient for achieving the desired functionality and device performance rather than automatically pursuing the most advanced node
- **Frugal manufacturing** — minimal tooling, energy, material and process steps

Examples of *Frugal Innovation*



Jaipur Foot is a landmark example of frugal innovation—a **low-cost, locally manufacturable prosthetic limb** designed for **affordability, durability and everyday mobility**, making **prosthetic care accessible to underserved populations**.

Examples of *Frugal Innovation*



Prof. Huang Shaoying's **AI-assisted low-cost MRI systems** at SUTD exemplify frugal innovation by simplifying MRI technology to create **more affordable, portable and accessible medical imaging solutions** without sacrificing essential diagnostic capability.

Frugal Innovation is a Multi-Objective Multi-Constraint Design Problem

- There is an interesting convergence happening between AI, sustainability, resource scarcity, circular economy, Design-for-X and frugal engineering.
- Imagine if you were given an existing product and challenged to achieve, say: 50% lower cost + 30% fewer parts + 30% lower energy/material use, while maintaining $\geq 90\%$ of essential functionality and required reliability. That is **FRUGAL INNOVATION!**
- **DESIGN.AI Perspective** - As frugality is a multi-objective engineering-design problem, it naturally makes modern AI-assisted optimization, generative design, digital twins and surrogate models particularly interesting tools for it to be achieved successfully.

$$\min \begin{bmatrix} \text{Cost} \\ \text{Material} \\ \text{Energy} \\ \text{Carbon} \\ \text{Complexity} \end{bmatrix} \quad \max \begin{bmatrix} \text{Performance} \\ \text{Reliability} \\ \text{Accessibility} \\ \text{Lifetime} \end{bmatrix}$$

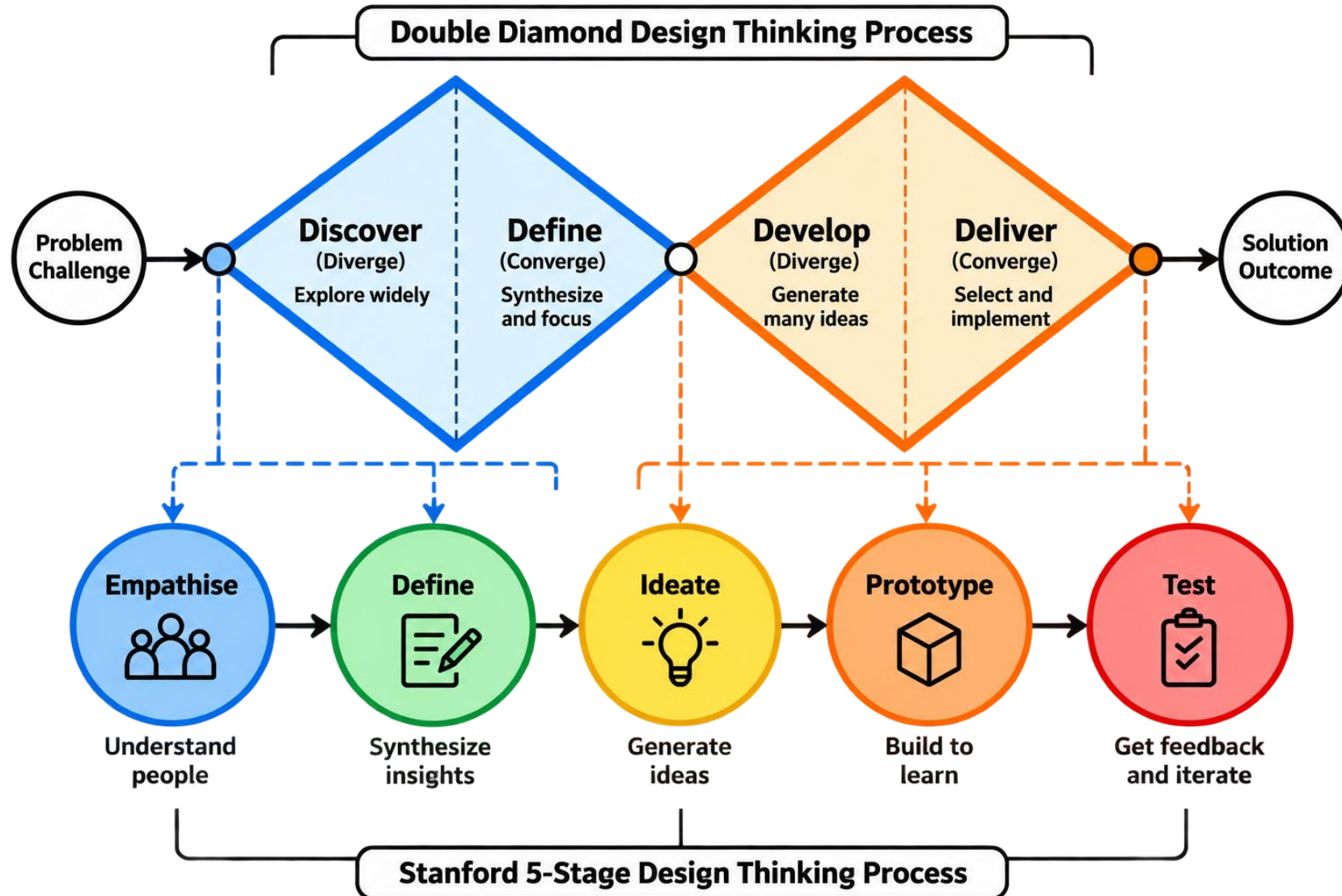
subject to safety, manufacturability and user requirements.

Frugal Value Index – An Intuitive Mathematical Formulation

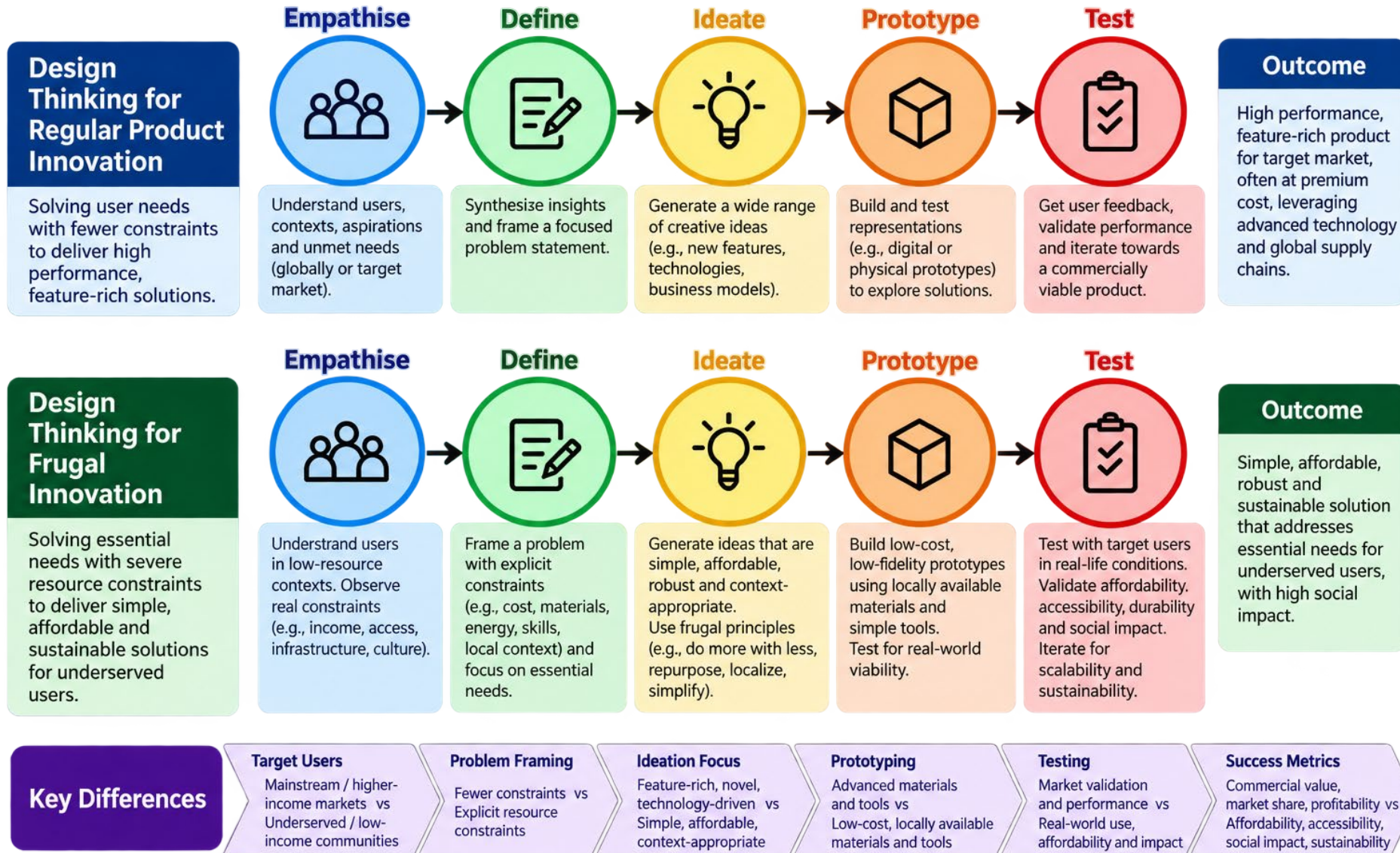
$$FVI = \frac{\text{Customer Value} \times \text{Performance} \times \text{Accessibility} \times \text{Robustness}}{\text{Cost} \times \text{Material} \times \text{Energy} \times \text{Complexity}}$$

The **Frugal Value Index (FVI)** expresses the principle of “**doing better with less.**” A strong frugal innovation maximizes **customer value, performance, accessibility, and robustness**, while minimizing the **cost, material use, energy consumption, and complexity** required to deliver that value. Thus, a **higher FVI indicates a more effective frugal solution—**provided essential functionality and safety are

Two Popular Design Thinking Frameworks – Same Message!



Design Thinking (with Frugal Innovation Mindset)



Think SMALL, Not BIG!!!

Think SMALL. Create BIG Impact.

Frugal innovation often comes from a small component, a simple attachment, or a clever tweak — **not a whole new system.**

Don't just think BIG...

A whole new system can be expensive, complex and hard to implement.



Entire system → Higher cost, longer development, harder to deploy

Think SMALL!

A simple, affordable component can solve a real user need and create big value.



Just a small component, but a big difference!

Specific component → Lower cost, faster to develop, easier to adopt, real impact

Everyday Examples of SMALL Components with BIG Impact

Simple ideas.
Real problems.
Meaningful impact.

1 Outdoor Trash Picker



A simple grabber to pick up litter without bending down.

Cleaner environment and safer, easier use

2 Fixture for Handheld Fan



A simple clip to attach a handheld fan to a shirt or cap.

Hands-free cooling in hot weather

3 Fully Foldable Shopping Trolley



A compact, fully foldable trolley for easy storage.

Convenient, space-saving and affordable

4 Smart Bicycle Lock



A smart lock with phone access, anti-theft alerts and simple design.

Enhanced security and peace of mind

5 Trash Bin Compressor



A simple compressor attachment to reduce trash volume.

More capacity, fewer bin changes

6 Smart Baby Monitor Wearable



A lightweight wearable to monitor baby's temperature, heart rate and sleep.

Peace of mind for parents

7 Multifunctional Stroller Hook



A sturdy hook with multiple functions and a locking mechanism.

Carry more, with greater safety

8 Water Bottle Holder for Joggers



A lightweight holder to carry a water bottle comfortably while jogging.

Stay hydrated, hands-free

9 Manual Vegetable Chopper



A simple pull-string mechanism to chop vegetables quickly.

Affordable, easy to use and time-saving



Small Components. Big Possibilities.

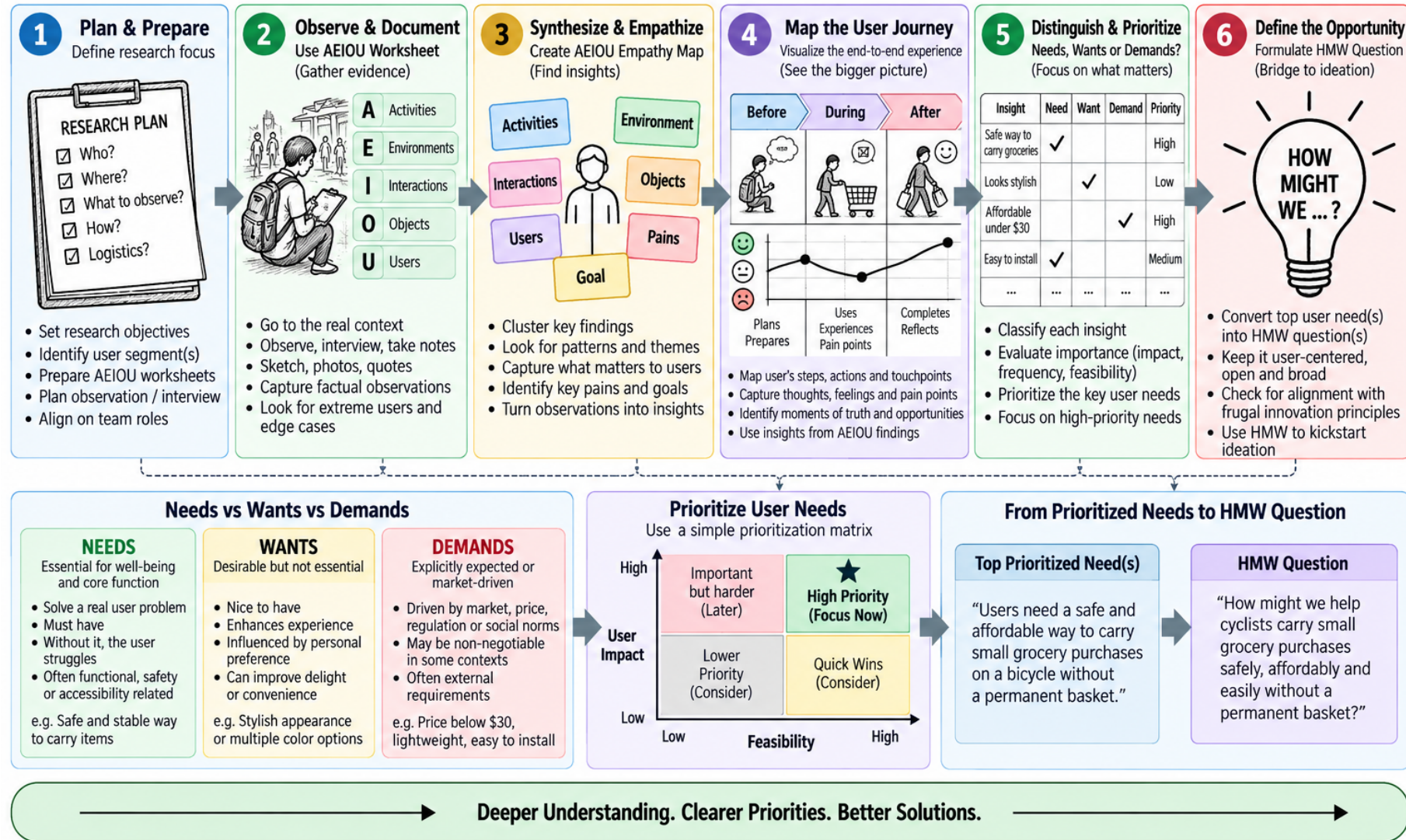
Look closely. Simplify. Reimagine.
Frugal innovation starts with a small idea.

A smaller idea can make a bigger, brighter tomorrow.

DISCOVER Phase of Design Thinking – The Workflow

From Observation to Opportunity

Understand Users → Find Insights → Prioritize Needs → Frame the Right HMW Question



Step-by-Step Workflow in DISCOVER Phase (1/4)

From Observation to Opportunity: Workflow to Frame Your HMW Question

1. Plan & Prepare — Decide what you want to understand

1. Define the **user group and context** you want to investigate.
2. Decide **who, where, what and how** you will observe.
3. Prepare your **AEIOU observation worksheet** and interview/observation plan.
4. At this stage, **do not start designing solutions**.

2. Observe & Document — Collect evidence using AEIOU

1. Go into the **real user context** wherever possible.
2. Observe and document **Activities, Environments, Interactions, Objects and Users (AEIOU)**.
3. Use **notes, sketches, photographs, quotations and interviews** as evidence.
4. Record **what you actually observe**, rather than what you assume users do.
5. Look for unusual situations, workarounds, frustrations and extreme users.

3. Synthesize & Empathize — Turn observations into insights

1. Consolidate your AEIOU observations into the **AEIOU Empathy Map**.
2. Cluster observations and look for **recurring patterns and themes**.
3. Identify important **pains, frustrations, workarounds and user goals**.
4. Ask: *“What really matters to this user, and why?”*
5. Convert individual observations into meaningful **user insights**.

Step-by-Step Workflow in DISCOVER Phase (2/4)

4. Map the User Journey — Understand the experience over time

1. Map the user's experience **before** → **during** → **after** the activity.
2. Identify the user's **actions, touchpoints, thoughts and feelings** at each stage.
3. Highlight **pain points, moments of truth and opportunity areas** along the journey.
4. Use evidence from your AEIOU research rather than inventing the journey.
5. Ask: “*At which point in this journey could a small intervention create significant value?*”

5. Distinguish Needs, Wants and Demands — Do not treat everything as equally important!

1. **Needs:** Essential requirements that solve a genuine user problem, e.g., *a safe and stable way to carry groceries on a bicycle.*
2. **Wants:** Desirable features that improve experience but are not essential, e.g., *stylish appearance or multiple colours.*
3. **Demands:** Explicit expectations or externally imposed requirements, often related to **price, regulations, market expectations or constraints**, e.g., *price below S\$30 or easy installation.*
4. Be careful not to mistake an attractive **want** for an important **need**.

Step-by-Step Workflow in DISCOVER Phase (3/4)

6. Prioritize the User Needs — Decide what deserves attention

1. Do **not** attempt to solve every pain point discovered during research.
2. Evaluate potential needs based on factors such as **user impact, importance/frequency and feasibility**.
3. A simple **User Impact × Feasibility matrix** can help.
4. Focus primarily on **high-impact, reasonably feasible needs**.
5. Remember your frugal mindset: look for opportunities where a **small intervention can create disproportionately large value**.

7. Formulate the Prioritized Need — State the problem without prescribing the solution

1. Express the most important finding from the user's perspective.
2. For example: *“Users need a safe and affordable way to carry small grocery purchases on a bicycle without a permanent basket.”*
3. Avoid prematurely specifying technologies or products such as *“We need to design a Bluetooth-enabled smart bicycle basket.”*
4. **Stay in the problem space before entering the solution space.**

Step-by-Step Workflow in DISCOVER Phase (4/4)

8. Frame the HMW Question — Convert the prioritized need into an innovation opportunity

1. Transform the prioritized need into an open-ended “**How Might We...?**” question.
2. For example: “*How might we help cyclists carry small grocery purchases safely, affordably and easily without a permanent basket?*”
3. A good HMW question should be **user-centred, open enough for multiple solutions, but focused enough to guide ideation**.
4. Do not embed your preferred solution inside the question.

9. Check the HMW Question through the Frugal Innovation Lens

1. Before ideating, ask whether the opportunity could potentially be addressed by **removing, reducing, combining, reusing, recycling or localizing**.
2. Ask: “*Can we solve this without redesigning the entire product or system?*”
3. Look particularly for **small components, attachments, accessories, modifications or services** that work with products users already own.

10. Only Now Enter Ideation

1. Once the HMW question is well framed, begin generating alternative concepts.
2. Do not become attached to the first solution.
3. Explore **multiple ways of answering the same HMW question**, including mechanical, digital and hybrid solutions.



Tips and Tricks

1. If we ask ChatGPT to generate a refined HMW, it will respond with something that is overly generalised.



To improve upon this...

2. Try to ask ChatGPT to generate needs or personas, then feed those needs and detailed personas into the prompt afterwards. The refined HMWs will be more meaningful and precise.

Example

HMW

1. *“HMW: develop a next generation EV” (initial prompt)*

NEED

2. *“What are some potential user needs for an EV?”*



ChatGPT Output

1. Performance

Extended range (400+ miles).

Fast charging (10-20 minutes for 80%).

Enhanced energy efficiency and regenerative braking.

2. Cost & Sustainability

Affordable pricing and maintenance.

Long-lasting, eco-friendly batteries (10+ years, recyclable materials).

Lower lifecycle emissions.

3. Charging & Infrastructure

Wide, reliable charging networks.

Smart and wireless charging options.

Vehicle-to-grid (V2G) energy integration.

3. Refined HMW

What?

“HMW design **electric vehicle systems** that seamlessly integrate high performance, sustainability, and user-centric features to re-define the **driving experience for urban citizens in diverse and challenging environments?**”

What does success look like?

For whom?

Why this HMW works:

- 1.Focuses on integration:** Encourages students to think holistically about combining performance, sustainability, and user experience.
- 2.Allows broad exploration:** "Redefine the driving experience" leaves room for creative interpretation and a range of innovative ideas, from user interfaces to modular systems.
- 3.Challenges conventional thinking:** "Diverse and challenging environments" inspires solutions that go beyond standard urban or highway settings, potentially sparking ideas for adaptability and inclusivity.
- 4.Feasible within 13 weeks:** Students can prototype specific subsystems or conceptualize bold solutions while grounded in real-world constraints.
If you'd like a more specific focus (e.g., battery systems, charging, or connectivity), I can refine it further.

Example

HMW

1. *“HMW: develop a next generation EV” (initial prompt)*

NEED

2. *“What are some potential personas for an EV?”*



ChatGPT Output

1. Alex, City Professional (28)
2. Maria, Suburban Caregiver (35)
3. Priya, Tech Enthusiast (30)
4. Kevin, Value Seeker (45)
5. Sarah, Logistics Coordinator (50)
6. Jake, Outdoor Enthusiast (40)
7. Victor, Business Executive (55)
8. Liam, Emergency Responder (38)

HMW

4. Further refined HMW for one of the personas “Emergency Responder”

For Persona 8: Emergency Responder (Liam, 38) here’s a redesigned "How Might We" (HMW) tailored to their unique needs and challenges:

*What?**What does success look like?*

"How might we design high-performance electric vehicles that ensure reliability, rapid response, and operational efficiency for emergency responders in extreme and unpredictable conditions?"

For whom?

Why this HMW works:

- 1.Focus on emergency needs:** Highlights the critical role of speed, reliability, and efficiency in emergency scenarios.
- 2.Challenges conventional EV design:** Encourages exploration of robust solutions for harsh climates, terrains, and unpredictable situations.
- 3.Room for innovation:** Opens possibilities for features like mobile energy storage, rapid charging, and specialized equipment integration.
- 4.Emphasizes practical use:** Grounded in operational realities like uptime, durability, and rapid deployment.

Affinity Analysis – Clustering and prioritizing Stakeholders' Needs

Best Practices

- **Collect needs holistically.**

Conduct this method with a cross-functional team, including stakeholders.

- **Cluster needs intuitively.**

Do not agonise over perfectly clustering the needs as themes will emerge organically.

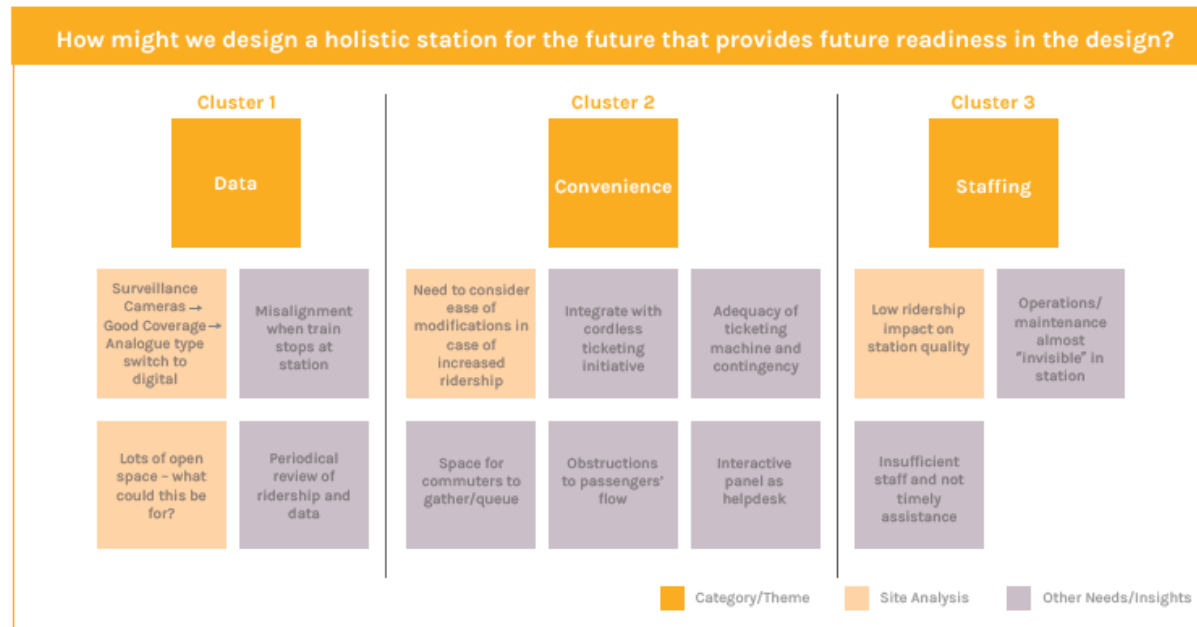
- **Label your clusters.**

Define and name themes based on content of ideas.

- **Discussion helps.**

Read needs aloud to the team while placing card on the wall, one at a time.

Worked Example



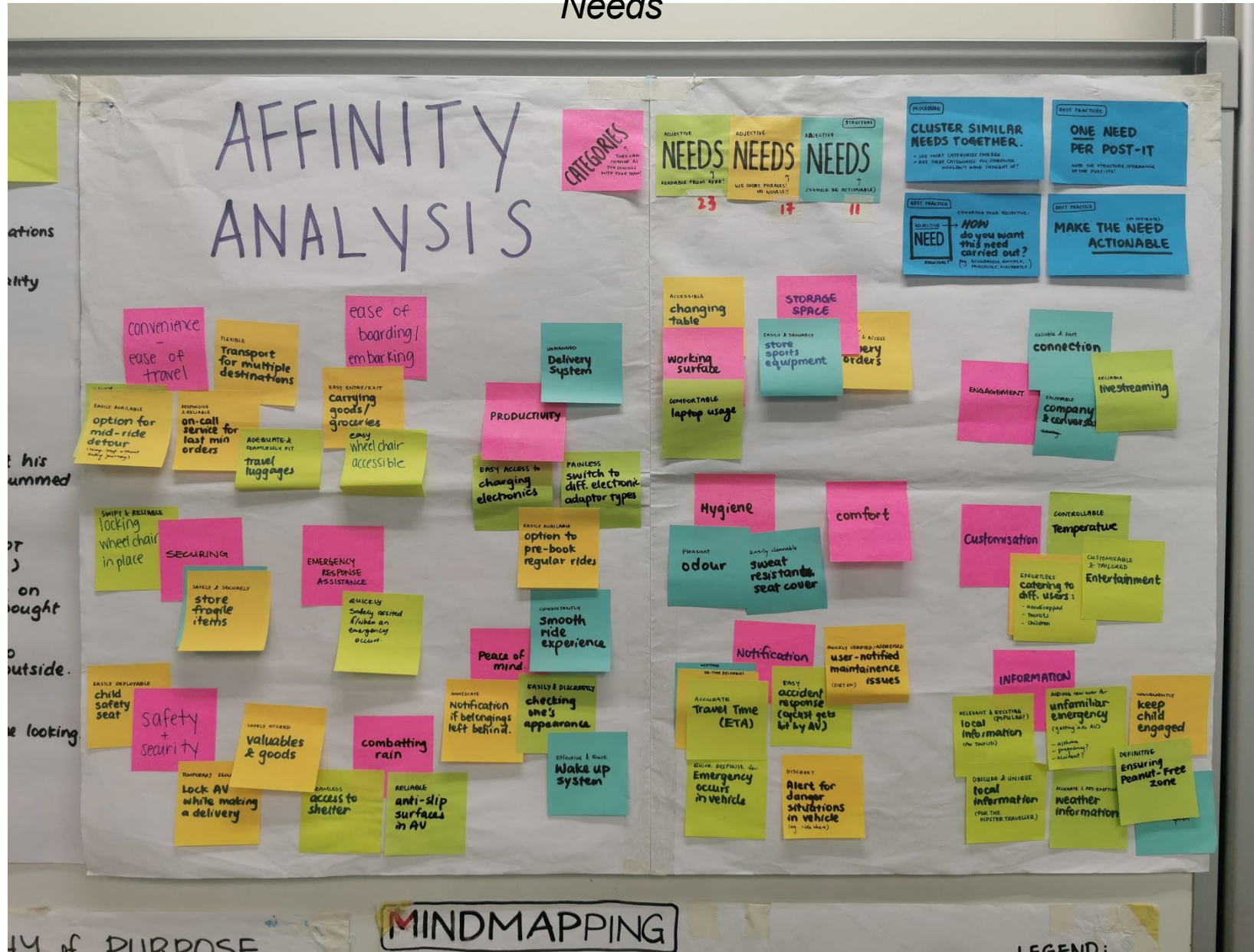
You may use miro board or post-its

Affinity Analysis – Clustering and prioritizing Stakeholders' Needs

How might we design a collaborative web platform around sharing, visualising, and comparing data for the future of young professionals and potential organisations for employment?



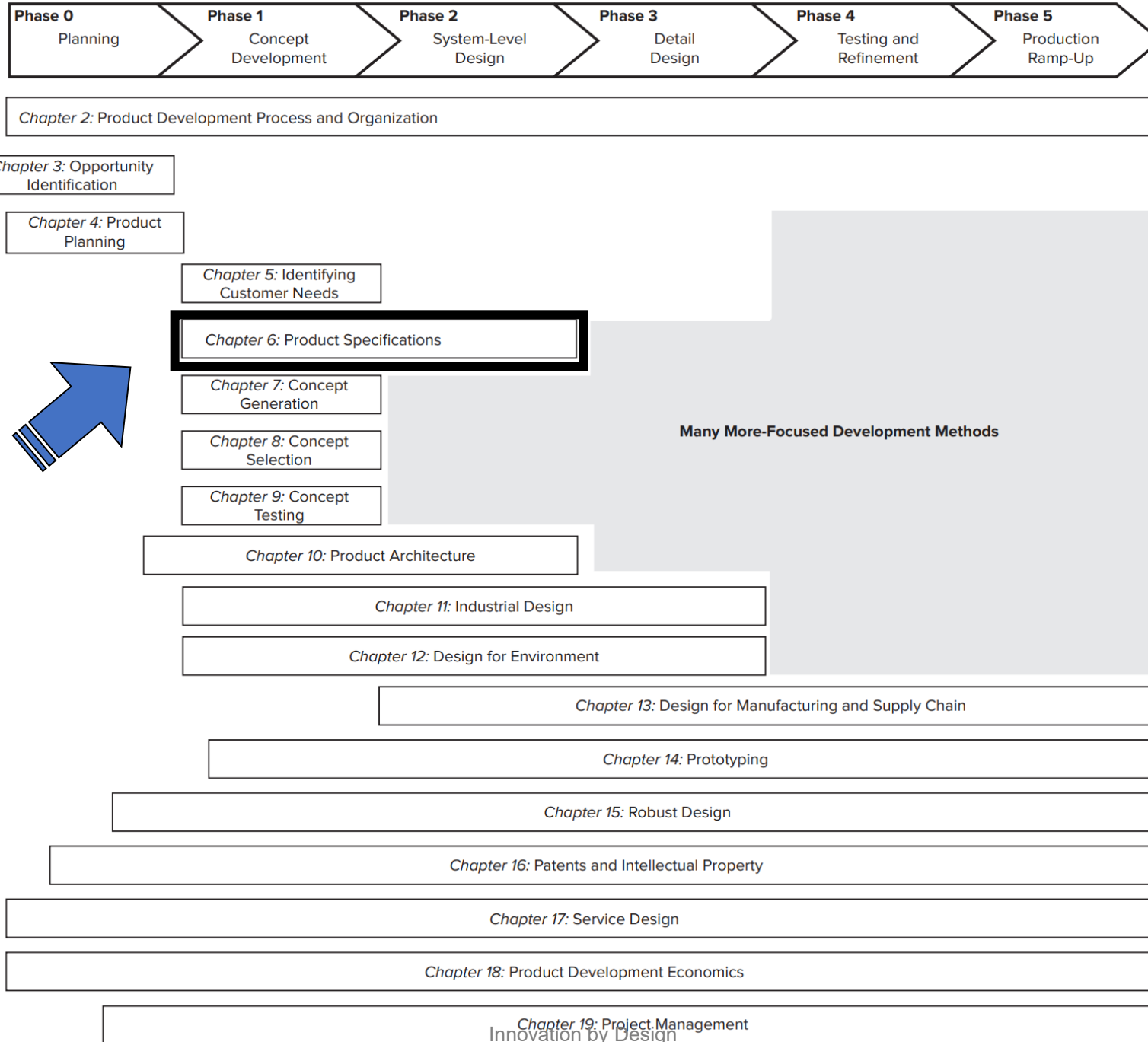
Affinity Analysis – Clustering and prioritizing Stakeholders' Needs



You may use miro board or post-its

Affinity Analysis – Clustering and prioritizing Stakeholders' Needs

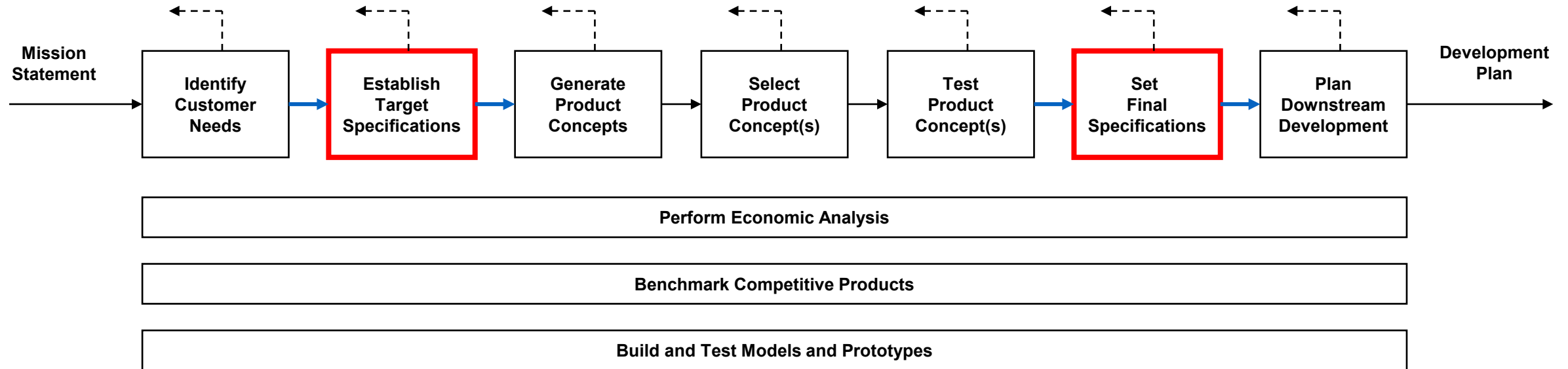




What are specifications?

- Specifications spell out in precise, measurable detail what the product has to do.
- Specifications represent an unambiguous agreement on what the team will attempt to achieve in order to satisfy the identified customer needs.
- **Must** include regulatory and agency approval requirements

Concept Development Process



Target Specs

Based on customer needs and benchmarking

Final Specs

Based on selected concept, feasibility, models, testing, and trade-offs

Product specifications

- Other names for ‘product specifications’
 - Product requirements
 - Technical specifications

Product Specifications Example: Mountain Bike Suspension Fork



A specification

- A ‘specification’ (*sing.*) consists of a metric and a value

Example:

metric → “Average time to assemble”

value → “less than 75 seconds”



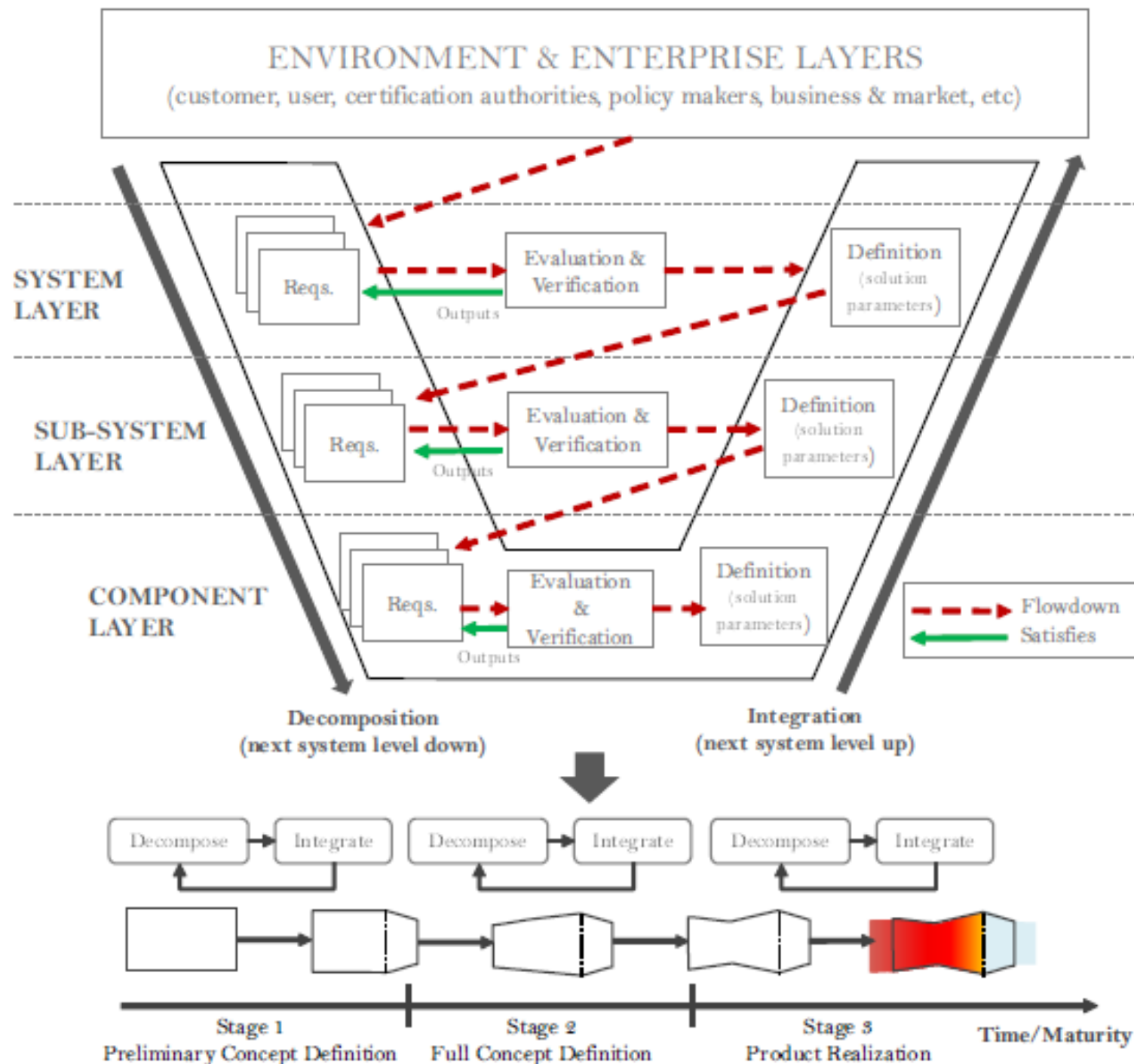
When are specs established?

- For simple products (e.g., soap):
early in the development process, right after identifying customer needs
- For technology-intensive products:
at least twice ...



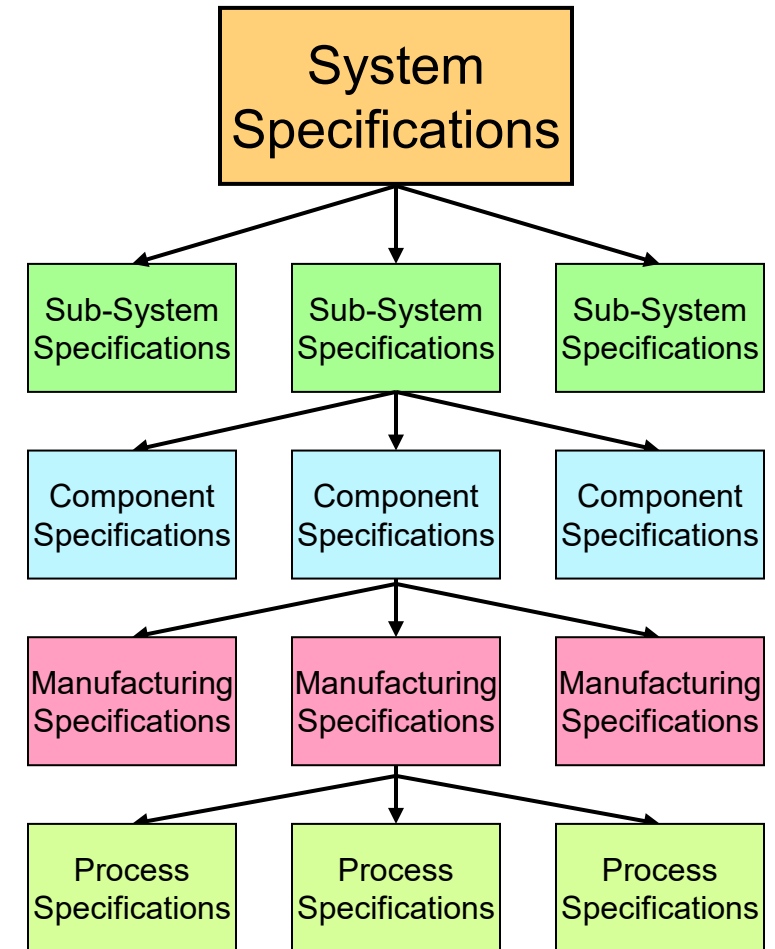
And for *REALLY* complex products

- Fernandes, J., Henriques, E., Silva, A., Pimentel, C., *Modelling the dynamics of complex early design processes: an agent-based approach*. Design Science, 3 (2017) e19. [DOI](#)
- Fernandes, J., Silva, A., Henriques, E., Moss, M. A., *Requirements change in complex technical systems: an empirical study of root causes*. Research in Engineering Design 26:1 (2015) 37-55. [DOI](#)
- Fernandes, J., Silva, A., Henriques, E., Moss, M. A., *A method for imprecision management in complex product development*. Research in Engineering Design 25:4 (2014) 309-324. [DOI](#)



Requirements Flow Down

- System-level specs determine whether we can meet the customer needs.
- Sub-system specs determine how to meet the system specs.
- Component specs determine how to meet the sub-system specs.
- Manufacturing specs determine how to meet the component specs.
- Process specs determine how to meet the component specs.



For technology-intensive products

- Target specifications
(representing hopes and aspirations)
are set immediately after the team
has identified the customer needs it
aims to meet.
- Refined specifications
(i.e., the specs the team aims to
achieve) are documented in the
project's "contract book".

Set Target Specifications

- ☐ Based on customer needs and benchmarks
- ☐ Develop metrics for each need
- ☐ Set ideal and acceptable values

Refine Specifications

- ☐ Based on selected concept and feasibility testing
- ☐ Technical modeling
- ☐ Trade-offs are critical

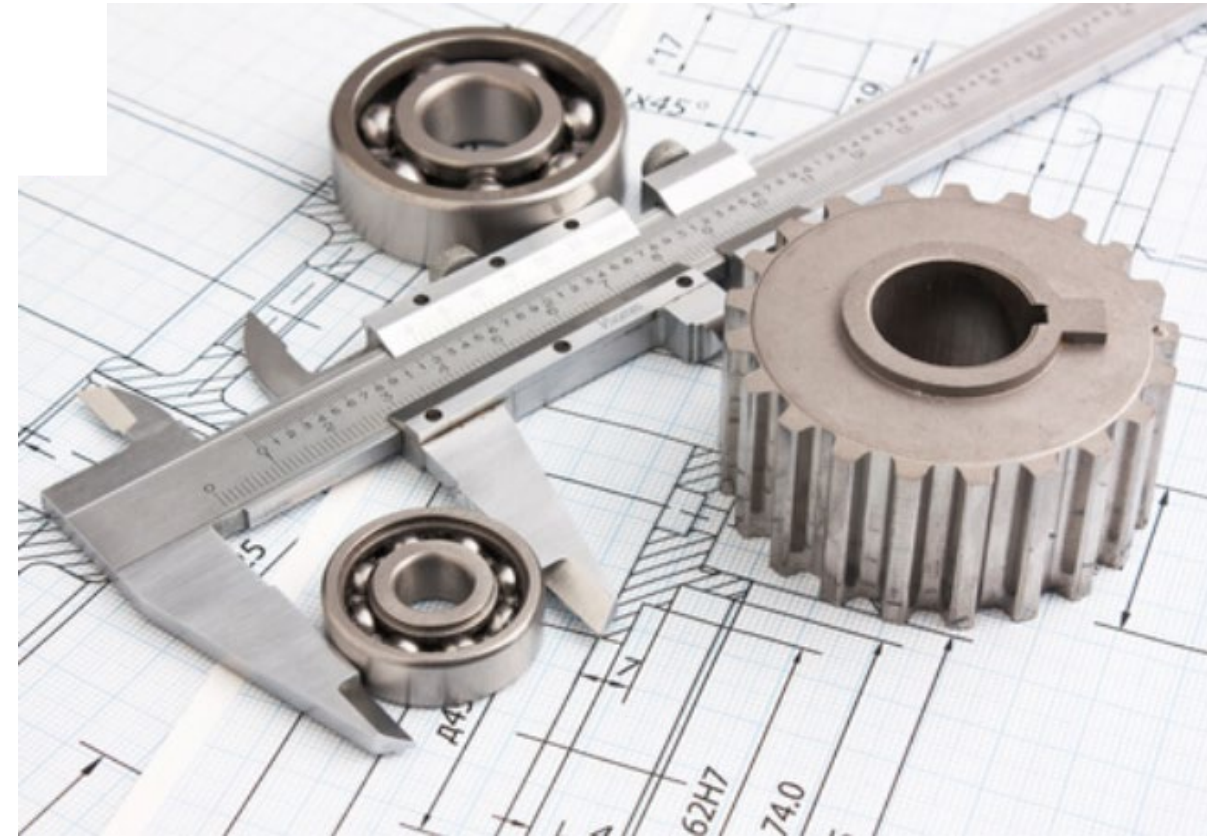
Target specifications

To establish target specs:

- Prepare list of metrics, using the needs/metrics matrix.
- Collect benchmarking information.
- Set ideal and marginally acceptable target values for each metric.
- Reflect on results and process.

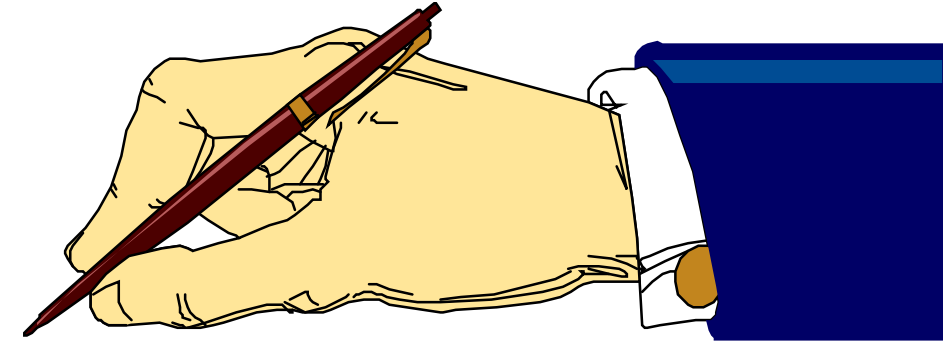
- Prepare a list of metrics

The underlying assumption is that consumer needs can be translated into precise, measurable specs and that meeting specs will result in satisfaction of the associated customer needs.



Metrics Exercise: Ball Point Pen

- Customer Need:
 - The pen writes smoothly.
- Metric(s):
 1. Variation in line thickness (mm)
 2. Variation in ink coverage (cc/mm^2)
 3. Functional range of writing force (N)
 4. Functional range of writing velocity (mm/sec)
 5. Functional range of pen angle from vertical (deg)
 6. Variation in resistance to translational motion (N)



Assuming that smooth writing can be characterized by:

- Good quality line
- Preservation of line quality
- Ease of use...

From Affinity Diagram Ranking Needs of Product (Cordless Screwdriver)

Cordless Screwdriver Survey

For each of the following cordless screwdriver features, please indicate on a scale of 1 to 5 how important the feature is to you. Please use the following scale:

1. Feature is undesirable. I would not consider a product with this feature.
2. Feature is not important, but I would not mind having it.
3. Feature would be nice to have, but is not necessary.
4. Feature is highly desirable, but I would consider a product without it.
5. Feature is critical. I would not consider a product without this feature.

_____ The screwdriver can drive screws into hardwood.

_____ The screwdriver can turn phillips, torx, socket, and hex head screws.

_____ The screwdriver can access screws at the end of deep, narrow holes.

And so forth.

The SD provides plenty of power to drive screws.

- * The SD maintains power for several hours of heavy use.
- ** The SD can drive screws into hardwood.

The SD drives sheet metal screws into metal ductwork.

- *** The SD drives screws faster than by hand.

The SD makes it easy to start a screw.

- * The SD retains the screw before it is driven.
- !* The SD can be used to create a pilot hole.

The SD works with a variety of screws.

- ** The SD can turn philips, torx, socket, and hex head screws.
- ** The SD can turn many sizes of screws.

The SD can access most screws.

The SD can be maneuvered in tight areas.

- ** The SD can access screws at the end of deep, narrow holes.

The SD turns screws that are in poor condition.

The SD can be used to remove grease and dirt from screws.

The SD allows the user to work with painted screws.

The SD feels good in the user's hand

- *** The SD is comfortable when the user pushes on it.
- *** The SD is comfortable when the user resists twisting.
 - * The SD is balanced in the user's hand.
- ! The SD is equally easy to use in right or left hands.
 - The SD weight is just right
 - The SD is warm to touch in cold weather.
 - The SD remains comfortable when left in the sun.

The SD is easy to control while turning screws.

- *** The user can easily push on the SD.
- *** The user can easily resist the SD twisting.
 - The SD can be locked "on."
- ! ** The SD speed can be controlled by the user while turning a screw.
 - * The SD remains aligned with the screw head without slipping.
- ** The user can easily see where the screw is.
 - * The SD does not strip screw heads.
 - * The SD is easily reversible.

Start with the Customer Needs

No.		Need	Imp.
1	The suspension	reduces vibration to the hands.	3
2	The suspension	allows easy traversal of slow, difficult terrain.	2
3	The suspension	enables high-speed descents on bumpy trails.	5
4	The suspension	allows sensitivity adjustment.	3
5	The suspension	preserves the steering characteristics of the bike.	4
6	The suspension	remains rigid during hard cornering.	4
7	The suspension	is lightweight.	4
8	The suspension	provides stiff mounting points for the brakes.	2
9	The suspension	fits a wide variety of bikes, wheels, and tires.	5
10	The suspension	is easy to install.	1
11	The suspension	works with fenders.	1
12	The suspension	instills pride.	5

... but remember

- Metrics should be dependent, NOT independent, variables.
- Metrics should be practical.
- Some needs cannot be easily translated into quantifiable metrics (subjective needs).
- Metrics should include popular criteria used for ‘marketplace’ comparisons.

Establish Metrics and Units

Metric No.	Need Nos.	Metric	Units
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	dB
2	2, 6	Spring preload	N
3	1, 3	Maximum value from the Monster	g
4	1, 3	Minimum descent time on test track	s
5	4	Damping coefficient adjustment range	N-s/m
6	5	Maximum travel (26-in. wheel)	mm
7	5	Rake offset	mm
8	6	Lateral stiffness at the tip	kN/m
9	7	Total mass	kg
10	8	Lateral stiffness at brake pivots	kN/m
11	9	Headset sizes	in.
12	9	Steertube length	mm

Needs vs. Metrics

Need		Metric
1	Reduces vibration to the hands	• Attenuation from dropout to handlebar at 10 Hz
2	Allows easy traversal of slow, difficult terrain	• Spring preload
3	Enables high-speed descents on bumpy trails	• Maximum value from the Monster
4	Allows sensitivity adjustment	• Minimum descent time on test track
5	Preserves the steering characteristics of the bike	Damping coefficient adjustment range
6	Remains rigid during hard cornering	Maximum travel (26-in. wheel)
7	Is lightweight	Rake offset
8	Provides stiff mounting points for the brakes	Lateral stiffness at the tip
9	Fits a wide variety of bikes, wheels, and tires	Total mass
10	Is easy to install	Lateral stiffness at brake pivots
11	Works with fenders	Headset sizes
		Steertube length
		Wheel sizes
		Maximum tire width
		Time to assemble to frame
		Fender compatibility
		Installs pride
		Unit manufacturing cost
		Time in spray chamber without water entry
		Cycles in mud chamber without contamination
		Time to disassemble/assemble for maintenance
		Special tools required for maintenance
		UV test duration to degrade rubber parts
		Monster cycles to failure
		Japan Industrial Standards test
		Bending strength (frontal loading)

Benchmarking

No product development team can expect to succeed without 'benchmarking' the project against competing products

Warning: Data in competitors' catalogues and supporting literature may *not* be accurate...



Benchmark on Metrics

Metric No.	Need Nos.	Metric	Units	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21	Tonka Pro	Gunhill Head Shox
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	dB	8	15	10	15	9	13
2	2, 6	Spring preload	N	550	760	500	710	480	680
3	1, 3	Maximum value from the Monster	g	3.6	3.2	3.7	3.3	3.7	3.4
4	1, 3	Minimum descent time on test track	s	13	11.3	12.6	11.2	13.2	11
5	4	Damping coefficient adjustment range	N-s/m	0	0	0	200	0	0
6	5	Maximum travel (26-in. wheel)	mm	28	48	43	46	33	38
7	5	Rake offset	mm	41.5	39	38	38	43.2	39
8	6	Lateral stiffness at the tip	kN/m	59	110	85	85	65	130
9	7	Total mass	kg	1.409	1.385	1.409	1.364	1.222	1.100

Benchmark on Customer Needs

No.	Need	Imp.	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx Ti 21	Tonka Pro	Gunhill Head Shox
1	Reduces vibration to the hands	3	•	••••	••	•••••	••	•••
2	Allows easy traversal of slow, difficult terrain	2	••	••••	•••	•••••	•••	•••••
3	Enables high-speed descents on bumpy trails	5	•	•••••	••	•••••	••	•••
4	Allows sensitivity adjustment	3	•	••••	••	•••••	••	•••
5	Preserves the steering characteristics of the bike	4	••••	••	•	••	•••••	•••••
6	Remains rigid during hard cornering	4	•	•••	•	•••••	•	•••••
7	Is lightweight	4	•	•••	•	•••	••••	•••••
8	Provides stiff mounting points for the brakes	2	•	••••	•••	•••	•••••	••
9	Fits a wide variety of bikes, wheels, and tires	5	••••	•••••	•••	•••••	•••	•

Setting target values

Set ideal and marginally acceptable target values for each metric.

- At least X
- At most X
- Between X and Y
- Exactly X
- A set of discrete values

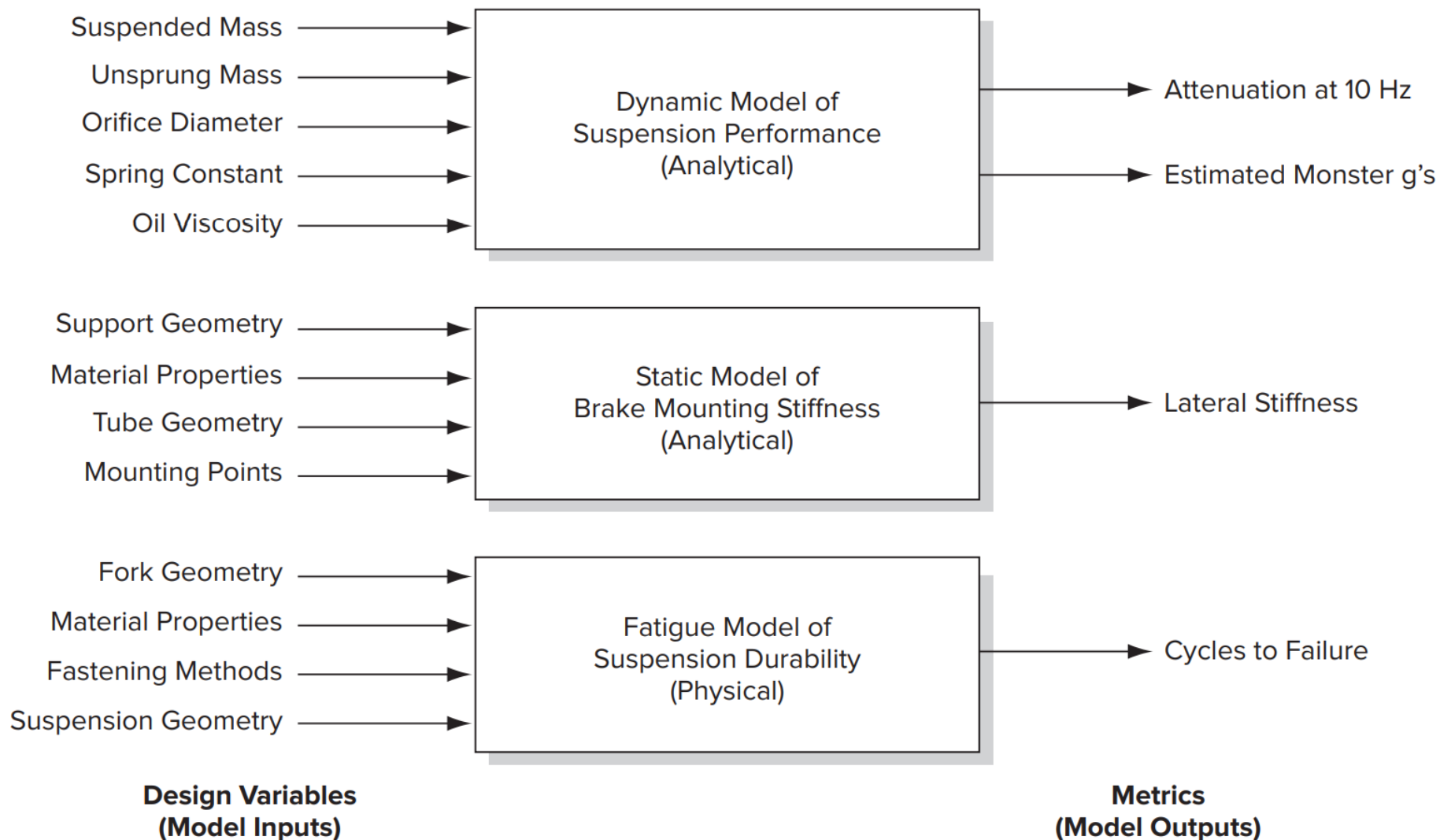
Assign Marginal and Ideal Values

Metric No.	Need Nos.	Metric	Units	Marginal Value	Ideal Value
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	dB	> 10	> 15
2	2, 6	Spring preload	N	480–800	650–700
3	1, 3	Maximum value from the Monster	g	< 3.5	< 3.2
4	1, 3	Minimum descent time on test track	s	< 13.0	< 11.0
5	4	Damping coefficient adjustment range	N-s/m	0	> 200
6	5	Maximum travel (26-in. wheel)	mm	33–50	45
7	5	Rake offset	mm	37–45	38
8	6	Lateral stiffness at the tip	kN/m	> 65	> 130
9	7	Total mass	kg	< 1.4	< 1.1
10	8	Lateral stiffness at brake pivots	kN/m	> 325	> 650

- Develop technical models of product
- Develop cost model
- Refine specs, making trade-offs where necessary
- Base trade-offs on tests using technical models and preliminarily cost model
- Reflect on results and process

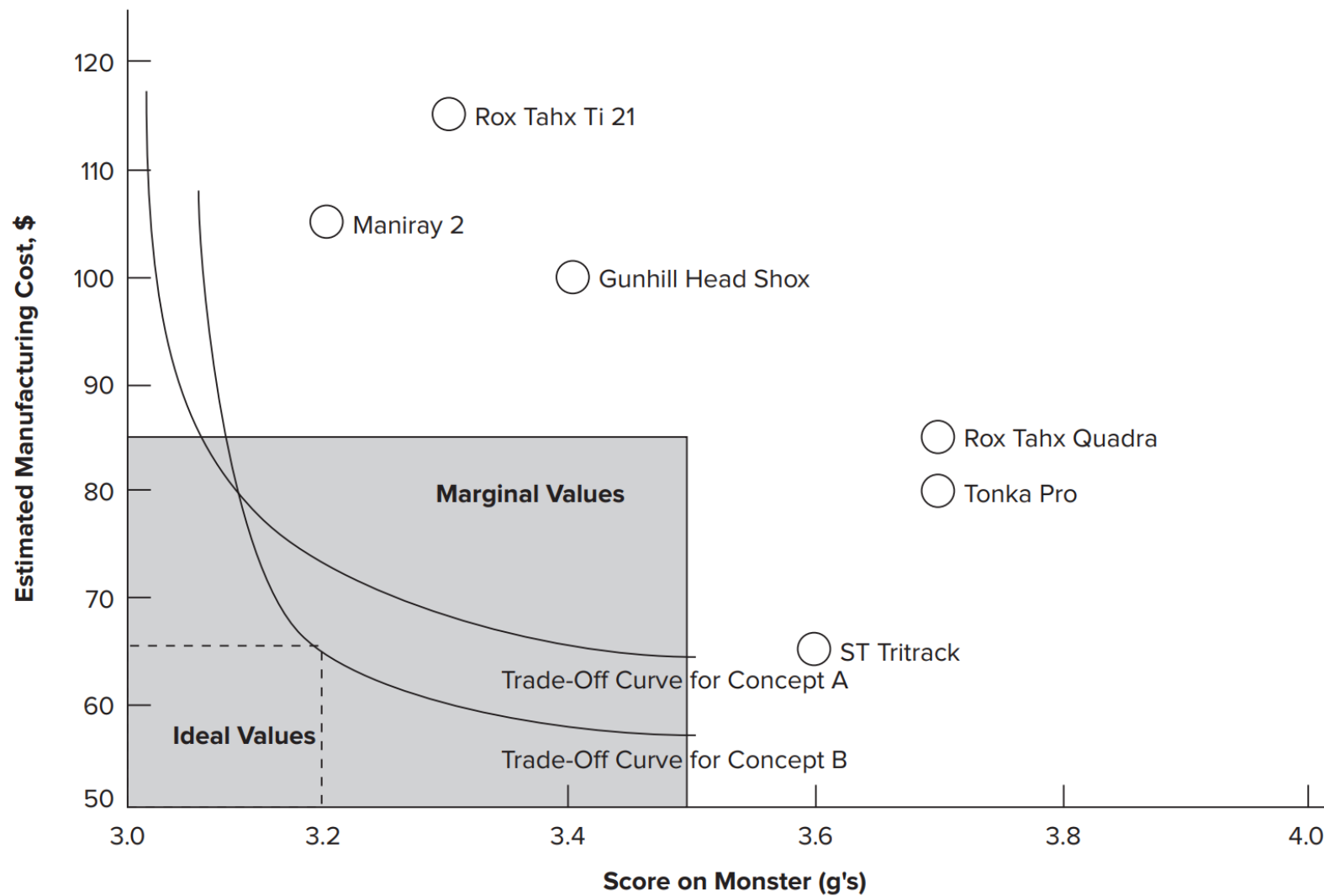
Develop technical models of the product:

- A technical model is a tool for predicting the values of different metrics for a particular set of design decisions.
- As used here, a ‘model’ may be an analytical or physical approximation of the product.

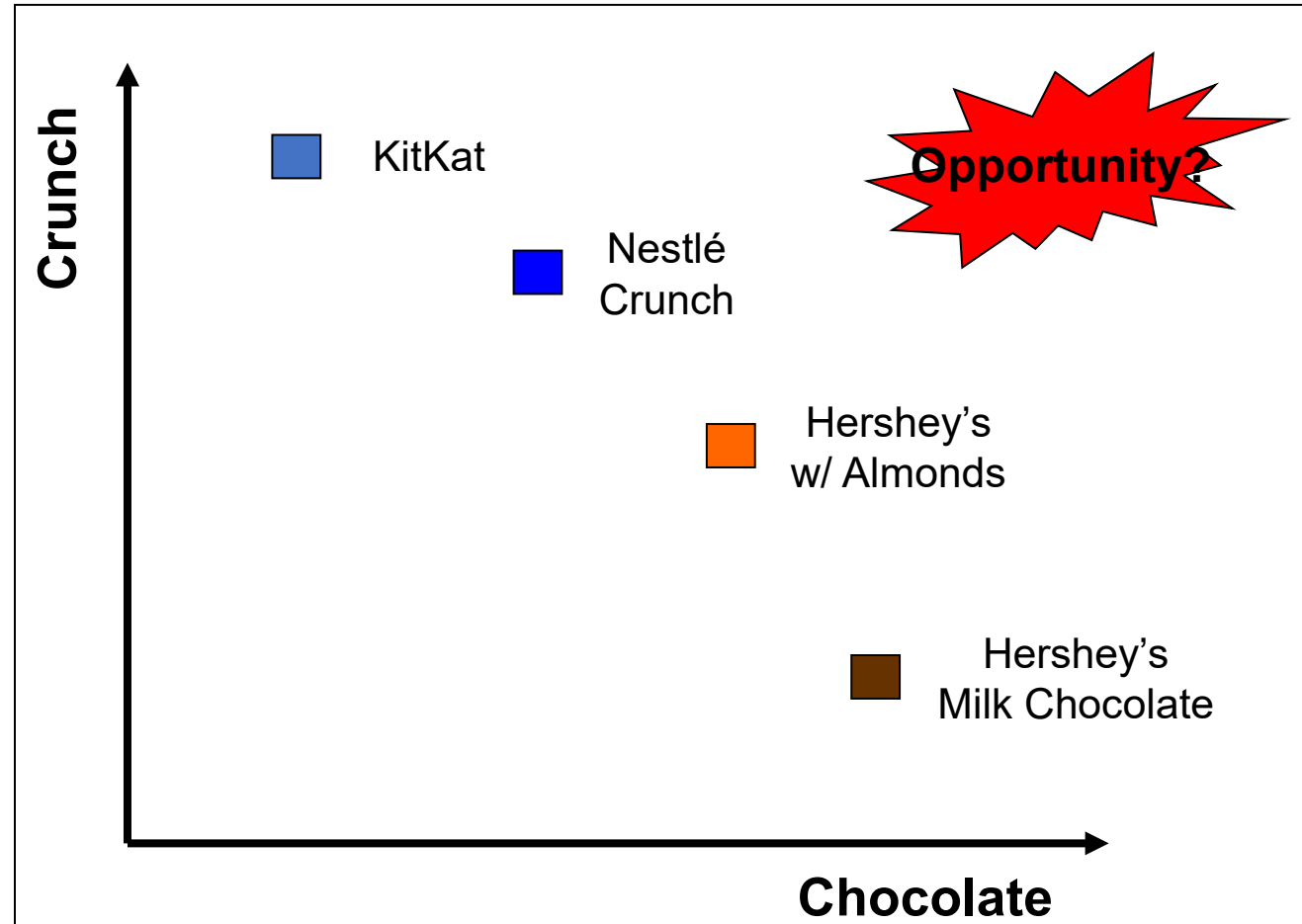


- How much uncertainty is there in the technical and cost models?
- Is the concept chosen by the team the best for that target market, or might it be more suitable for another market or segment (low-end or high-end instead of middle?)

Specification Trade-offs



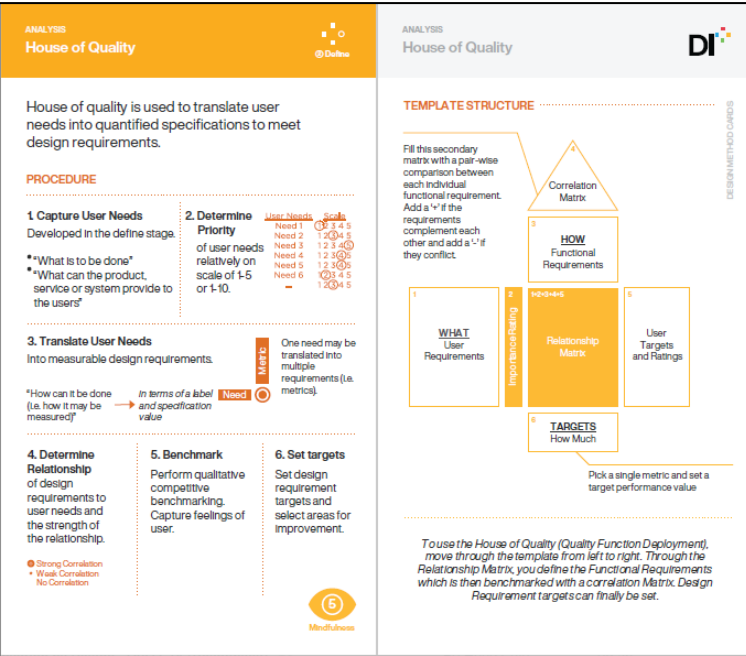
Perceptual Mapping Exercise



Set Final Specifications

No.	Metric	Unit	Value
1	Attenuation from dropout to handlebar at 10 Hz	dB	> 12
2	Spring preload	N	600–650
3	Maximum value from the Monster	g	< 3.4
4	Minimum descent time on test track	s	< 11.5
5	Damping coefficient adjustment range	N-s/m	> 100
6	Maximum travel (26-in. wheel)	mm	43
7	Rake offset	mm	38
8	Lateral stiffness at the tip	kN/m	> 75
9	Total mass	kg	< 1.4
10	Lateral stiffness at brake pivots	kN/m	> 425

Quality Function Deployment (House of Quality)

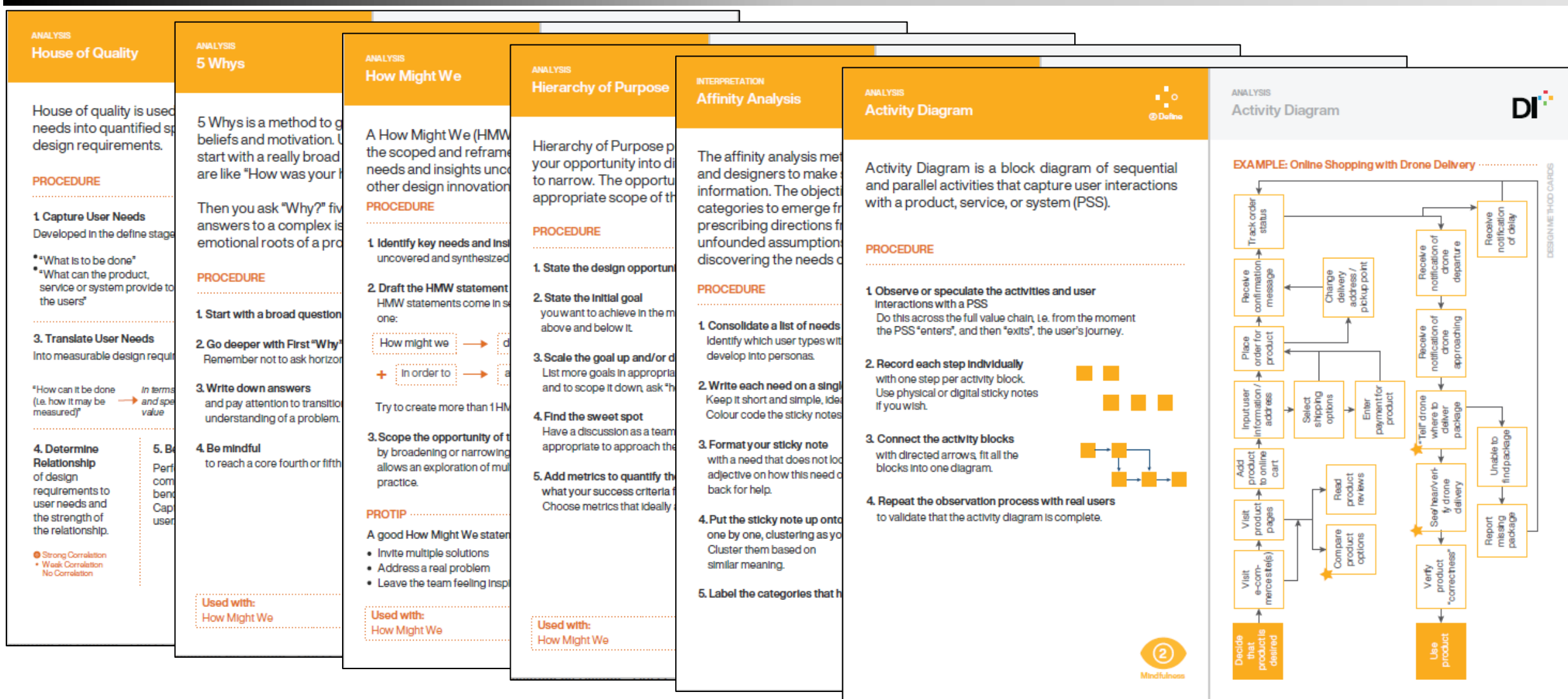


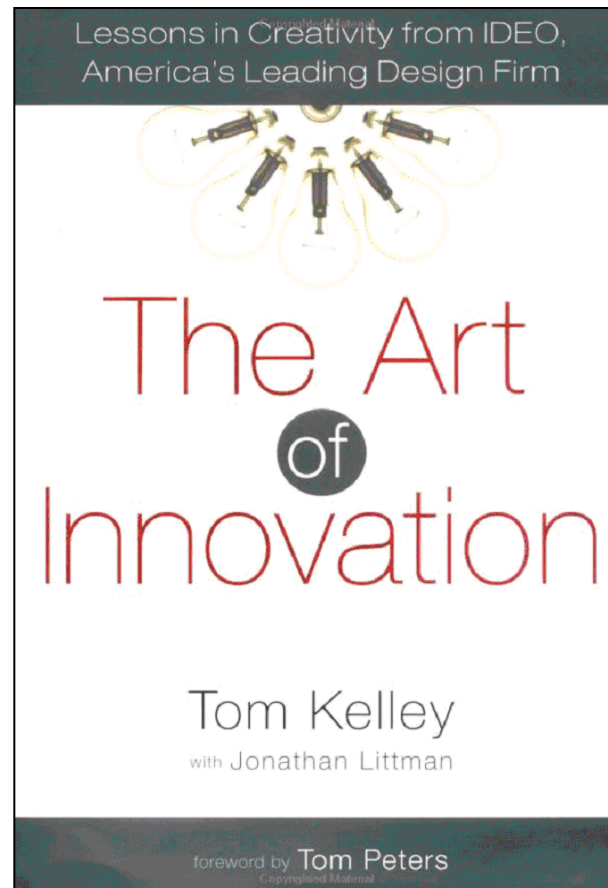
Maximize or Minimize												Comparative Assessment				
House of Quality FlyCo Drones												1	2	3	4	5
Customer Specifications / Importance																
1	2	3	4	5	6	7	8	9	10	11	12					
1	Sylish	2	1	1												
2	Lightweight	3	1	1												
3	Quiet	5														
4	Super HD Camera	5														
5	Stable	5														
6	Long Range	4														
7	Self-Directing	4														
8	Programmable Paths	5														
9	High-Quality Finish	2														
10	Durable Body	3														
11	Long Drone Battery Life	5														
12	Long Remote Battery Life	5														

Reflect on results and process

- Is the product a winner?
 - Product concept should allow team to see specs in a way that the price will meet the customer needs and excel competitively.
 - Product concept + specs should offer prospect of competitive pricing + performance meeting or exceeding customer needs
 - If not, return to concept generation and selection phase or abandon project.

Other methods you can use:





“The Art of Innovation” Tom Kelley
Currency Doubleday 2001
ISBN 0-385-49984-1

You may start selling your concept/idea that specifically addresses needs of your users

Paul Buchheit points out that trying to sell something bad can be a source of better ideas:

"The best technique I've found for dealing with YC companies that have bad ideas is to tell them to go sell the product ASAP (before wasting time building it). Not only do they learn that nobody wants what they are building, they very often come back with a real idea that they discovered in the process of trying to sell the bad idea."

That's all Folks!